AoE2 Civ Explorer

Code documentation

What each part of the code does, and why

1 November 2026 · 8 chapters and 25 subchapters · 396 functions and 7 objects

This document explains the code of AoE2 Civ Explorer. Each folder is a chapter, and each file in it a subchapter. In the subchapters on scripts, each function defined at the top level of its script is a section, numbered after its chapter and subchapter, so the fourth section of the first file in chapter 4 is 4.1.4. A section gives the lines the function takes up, says what it does and why, and lists the functions that call it and the functions it calls. A function handed on to be called later, such as an event handler, counts as called by the function that hands it on. Helpers defined inside a function are part of that function's text.

"Exported" marks a function that other scripts import, and "internal" one that only its own script uses. The subchapters on the page, its style sheets and the data are split into parts in the same way, each with its lines. Rules and requirements are those of RULES.md and REQUIREMENTS.md.

Every name of a folder, a file or a function links to its chapter, subchapter or section, and the page header always names the chapter and the subchapter. The contents list the chapters, each with its subchapters indented under it. The four UML diagrams after the contents show how the parts and the scripts use each other, and the index at the end lists every chapter, subchapter and function in alphabetical order.

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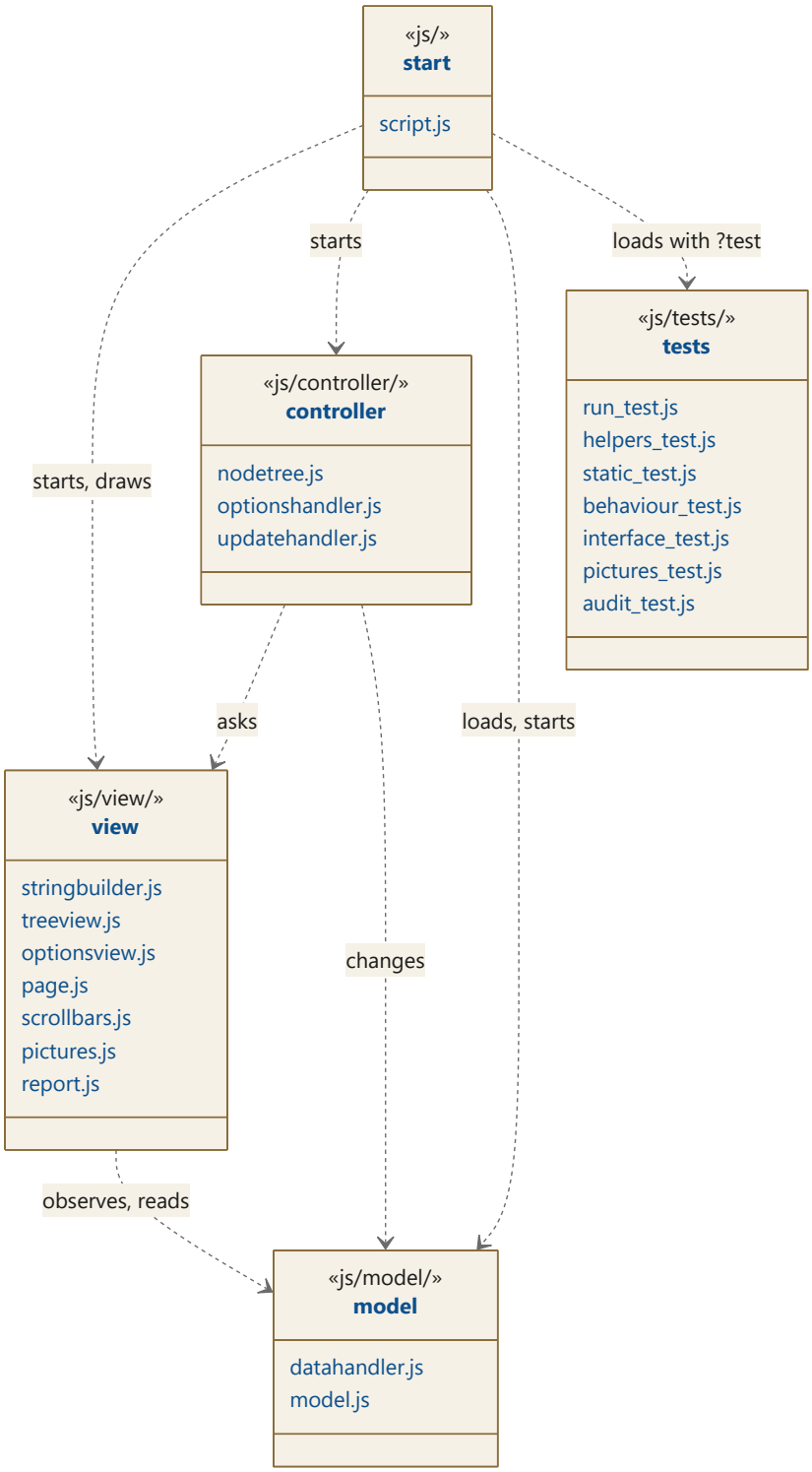
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UML diagrams

Four class diagrams show how the scripts use each other. The first, the one in MAINTAINER.md, shows the code in its three parts, model, view and controller, file by file; the other three show the scripts of each part, function by function. In each, an arrow points from what uses to what is used. No arrow crosses or touches another, or runs under a box or a label: making this document checks every diagram for it, and stops if one does. Each box and each name in a box links to its chapter, subchapter or section.

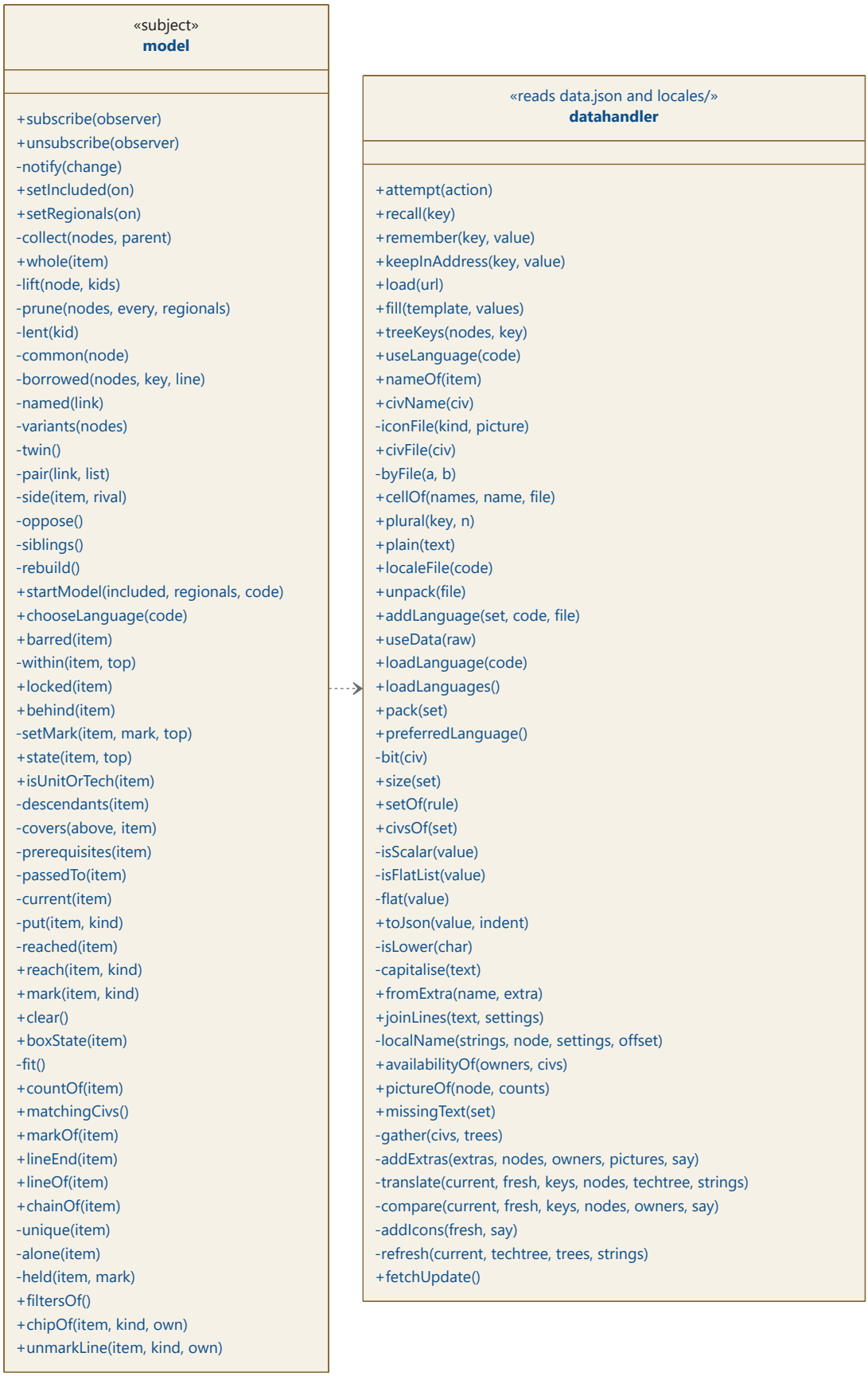
Overview

Each box is a folder, with the scripts in it; `start` is js/, whose script.js starts the page. Each arrow points from a part to a part whose scripts it uses, and says what for. The tests use all three parts as well; those arrows are left out, because with them some arrows would have to cross. A box links to its folder's chapter, and a script to its subchapter.



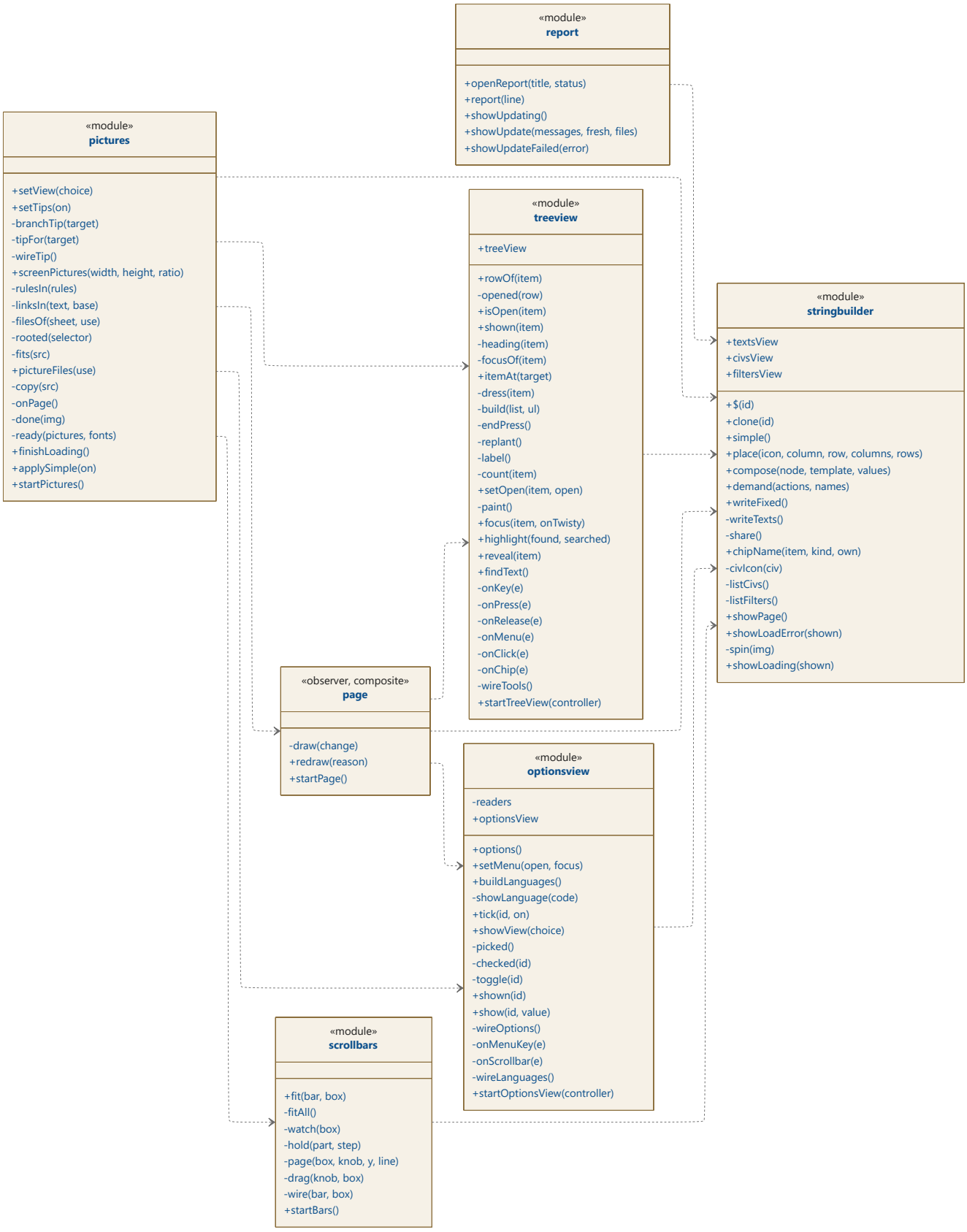
Model

The scripts of js/model/, each with its functions and objects: + marks what a script exports, and − what it keeps to itself. The line under a script's name gives its role: its part in model, view and controller, which MAINTAINER.md explains, what it reads, or "module". An arrow points from a script to a script of the same folder that it imports; what the scripts use in other folders is in the overview.



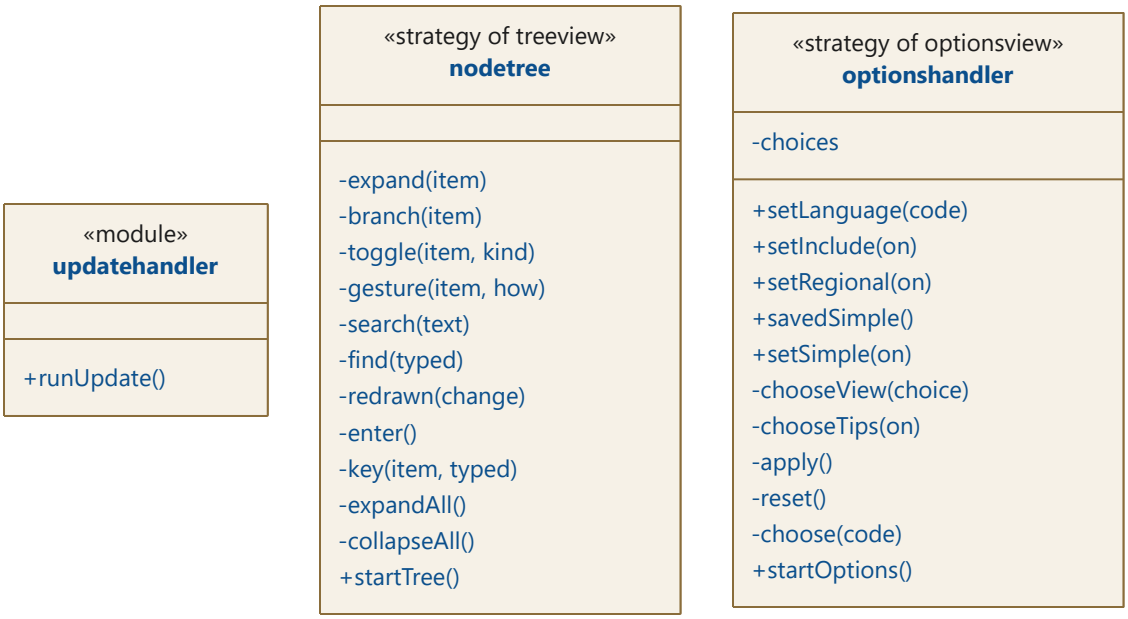
View

The scripts of js/view/, each with its functions and objects, drawn as in the diagram of the model: + for what a script exports, − for what it keeps to itself, its role under its name, and an arrow to each script of the folder that it imports.



Controller

The scripts of js/controller/, each with its functions and objects, drawn as in the diagram of the model. None of them imports another of the folder, so there are no arrows: what they use of the view and the model is in the overview.



1 Main folder

2 files 2377 lines

The page and its data: index.html, which the browser opens, and data.json, which the scripts load at start-up. The other files there, such as README.md, are not part of the code.

Subchapters: 1.1 index.html, 1.2 data.json

1.1 index.html

Folder: the main folder 105 lines 11 parts

Loads: css/base.css, css/pictures.css and js/script.js

The page's structure, and all of its HTML (Rule 2). It holds no text of its own: the scripts fill in every word in the chosen language, and copy its templates instead of writing HTML.

1.1.1 Head

lines 1–20

Sets the language, character set, viewport, light and dark colour schemes, title and description, and what link previews show: the `og:` tags give a title, a description, `assets/images/preview.jpg` and the page's address. It links base.css, pictures.css and script.js, as a module. It names no tab icon: browsers fetch favicon.ico from the main folder by themselves.

1.1.2 Loading screen

line 22

The screen shown without Simple UI until every picture has loaded, with its text. `showLoading()` adds the turning picture from its template.

1.1.3 Header

lines 23–26

Holds the title twice: the full sentence of Req. 2 for Simple UI, and "Civ Explorer" for the pictures (Req. 0). Then the cogwheel Options button (Req. 6), a text character with a hidden label. Hidden until the page starts.

1.1.4 Messages

lines 27–28

A noscript line for browsers without JavaScript, and a load error shown if data.json cannot be loaded, so a visitor is never left with a blank page.

1.1.5 Tree pane

lines 29–50

The start of the content, then the left pane, in the order of Req. 3: heading, Clear, Expand all and Collapse all, the Find field with its label, clearing cross and live match count, the must have and must lack legend, and the empty tree.

1.1.6 Civilisations pane

lines 51–64

The right pane (Req. 4): heading, live count, the Filters list (Req. 22), the civilisation list and the hint shown when it is empty. It scrolls on its own, so the page never does (Rule 12).

1.1.7 Footer

lines 65–66

Names and links the data source and its date (Req. 5), then Microsoft's notice, in English, with links to its rules and to the project. The code writes the sentence around the links.

1.1.8 Options window

lines 67–83

A dialog (Req. 7, 8, 32): title, the language label, button and list, then the settings: Simple UI, Include Unique and Universal with its description, Include Regionals, Civ view and Tooltips; and Cancel and OK.

1.1.9 Report window

lines 84–92

A dialog for the reports of the tests and the updater: title, live status, result list, download links and Close. Always in English, so it is marked `lang="en"`.

1.1.10 Templates

lines 93–102

Markup for tree rows, Filters buttons, civilisations, whose links open in a new tab, the loading picture, the icons of civilisations and of the tree, language options, report lines, downloads and the scroll bar. The code copies these instead of writing HTML (Rule 2).

1.1.11 Tooltip

line 103

The note that follows the mouse without Simple UI: over the tree, over every control, each naming its text with `data-tip`, and over a link, with its address. Hidden otherwise.

1.2 data.json

Folder: the main folder 2272 lines 12 keys

All of the page's data, in English: the civilisations, the tree, who has each item, its icon and the page's texts. `useData()` reads it at start-up, and the updater writes it through `pack()` and `toJson()`: short objects and lists on one line, the rest indented by two spaces. The other languages' texts are in `locales/`, one file each, laid out like the `overrides` and `text` here.

An item is named by its English name throughout: the tree, `availability`, `pictures`, `extras` and `text.items` use the same names as keys.

Lines	Key	What is there
2–11	`source`	Where the data comes from: aoe2techtree.net and its data address, `helpOffset` (taken from an item's help string number to find its text, in `localName()`), the month of the last update, and Microsoft's notice with the rules and the project it links.
12–30	`languages`	Every language by its code, with its name in itself, its language tag, and whether its words are spaced and its hyphens at line ends are merged. English comes first.
31	`civs`	The civilisations, in the order whose bits the sets use. A set is a number with one bit per civilisation (`bit()`).
32–46	`icons`	Where the pictures are: the file names of civilisations that differ from their names, and the folder of the tree's icons.
35–39	`groups`	The icon of each top-level group, by kind and picture.
40–45	`atlas`	For each kind, the pictures in its atlas, which `cellIn()` places by file name.
47–686	`tree`	The tree as nested objects: `{"group": …}` for a top-level group or a group in a building, `{"item": …}` for an item, with `kids` for what is under it and the flags `own`, `lines`, `excludes` and `replaces` where they belong. `rebuild()` reads it.
687–1158	`availability`	Who has each item: `"all"`, `{"has": […]}` or `{"lacks": […]}`, whichever is shortest (`availabilityOf()`). `setOf()` turns it into a set.
1159–1630	`pictures`	Each item's icon: its kind, its picture in that kind's atlas, and its reach, common, regional or unique, which Include Regionals and the outlines use.
1631	`ignore`	Names on aoe2techtree.net that the updater does not report as new items.
1632–1641	`extras`	Items aoe2techtree.net does not show in its tree, with their kind, picture, reach, name number and civilisations. `fromExtra()` turns each into availability and a picture.
1642	`overrides`	Names to use instead of the game's own, where it gets one wrong.
1643–1669	`tools`	The texts of the tests' report and of the updater, in English only, with `{}` placeholders that `fill()` fills in.
1670–2271	`text`	The page's texts in English, in four parts.
1671–1733	`ui`	The page's own texts, by the name `data-text`, or for a tooltip `data-tip`, gives in index.html, counted ones as `{"one": …, "other": …}` for `plural()`.
1734–1740	`groups`	The name of each group.
1741–1798	`civs`	The name of each civilisation.
1799–2270	`items`	The name of each item.

2 css/

2 files 462 lines

The two style sheets index.html links: base.css, for Simple UI and the layout either way, and pictures.css, for every rule with a picture or a font, which Simple UI turns off.

Subchapters: 2.1 base.css, 2.2 pictures.css

2.1 base.css

Folder: css/ 201 lines 12 parts
Loaded by: index.html

How the page looks with Simple UI, and its layout either way: colours, sizes and layouts for big and small screens, in light and dark mode. Every rule with a picture or a font is in pictures.css instead.

2.1.1 Colours — lines 1–33

Named colours for light mode, redefined for dark mode: text, lines, marks, hover, Find's yellow, the green and red of the marks, and the dark box behind the white dash; and the 44-pixel target size and the font. The tests measure their contrast.

2.1.2 Base — lines 34–46

Box sizing, `hidden` that always hides, a full-height page that never scrolls (Rule 12), the system font with 1.5 line spacing (WCAG 1.4.8), a 3 px focus outline (WCAG 2.4.13), and a class that hides text from sight but not from screen readers.

2.1.3 Header and language — lines 48–66

Lays out the title and cogwheel, and in the Options window the language label, its button with a turning chevron, and the list of languages, where the current one is marked with a bar and bold.

2.1.4 Panes — lines 68–83

Two panes side by side from 720 px wide; below that they share a phone's height, and the footer is hidden. Each pane's head stays in place while its list scrolls.

2.1.5 Controls — lines 84–101

Buttons and the Find field are at least 44 by 44 px (WCAG 2.5.5), with visible borders and a hover colour. Find's message keeps its column, so the box never changes size while typing (Rule 18). The legend, and the scrolling lists.

2.1.6 Tree — lines 103–121

Indented rows with dotted guide lines, 44 px rows, turning arrows, pressed and greyed-out rows, and found rows in yellow with the light colours (Req. 20) whatever the colour scheme.

2.1.7 Check boxes — lines 122–140

Marks drawn with borders (Req. 14, 23, 26): a green tick, a red cross, and a white dash on a dark box, in system colours when colours are forced.

2.1.8 Names and views — lines 141–145

Hides Civ view and Tooltips, which only the pictures use; makes top-level and group names bold; keeps counts on one line at the right; and draws focus rings inside rows.

2.1.9 Filters and list — lines 147–163

Filters as a wrapping row of rounded chips under the pane's head, the civilisation list in columns that never split a name, the empty hint and the footer.

2.1.10 Dialogs — lines 165–185

Size, padding, border and backdrop of the Options and report windows. The Options window is a grid of fixed height, so opening the language list never resizes it (Rule 18); only the report's list scrolls.

2.1.11 Small screens

lines 187–198

On short screens everything scrolls as one, as at 400% zoom (WCAG 1.4.10); on narrow ones the padding and the tree's indent shrink.

2.1.12 Test helper

lines 200–201

The wider text spacing the tests apply (WCAG 1.4.12). Used only while testing.

2.2 pictures.css

Folder: css/ 261 lines 16 parts

Loaded by: index.html

How the page looks without Simple UI. Every rule here that shows a picture starts with `.pictures` or `html.pictures`, so Simple UI turns them all off (Rules 9, 22), and picture addresses go from this folder, as `../assets/`. The `[data-assets="4k"]` and `[data-background]` rules pick the pictures for the screen.

2.2.1 Parchment and cursors

lines 1–23

The parchment behind the page, fitted so the content lies inside its frame on any window, in the four sizes; the sword and the pointing hand as cursors, the `_x2` ones drawn at two thirds of their size, at `1.5x` in `image-set`, so they are as large as the `_x1` ones.

2.2.2 Header and footer

lines 24–46

The footer's colours, the title in its frame with the short title, the load error, and the Options button from its three-state sheet.

2.2.3 Fonts

lines 47–56

Georgia, Times New Roman Bold and Trajan Pro Bold from `assets/fonts/`, under their own names, trying the installed copy first. Only rules for the pictures use them, so Simple UI never downloads a font.

2.2.4 Text buttons

lines 57–66

Every button with words, but those of the report and the language box, shows the button picture, normal, under the mouse or pressed, with its ends drawn apart so it can be any width.

2.2.5 Loading screen

lines 68–71

The screen that covers the page while the pictures load, and its turning picture, the only animation, which stops with reduced motion (Rule 11).

2.2.6 Boxes and radio buttons

lines 73–89

Check boxes from their sheet, a row for each state and a column for normal, under the mouse and pressed, with a black dash; Tooltips and the Civ view radio buttons, shown only here.

2.2.7 Frames

lines 90–97

The 9-slice frames of the two panes, and their fonts and colours.

2.2.8 Options window

lines 98–113

The window's frame, measured to fit around the same window as with Simple UI, with its contents 60 pixels from the frame's sides, or 120 of the screen's own pixels on a 4K screen (`--pixel`), and the settings lit under the mouse.

2.2.9 Language box

lines 114–145

The language menu as the game's combo box: its picture, 219 by 35 pixels at the right of its row, with the language's name and, at its right end, the arrow from its atlas, pointing down while the list is closed and up while it is open, normal, under the mouse or pressed, and 35 by 28 of the screen's own pixels on a 4K screen. The list lies over the settings below the box, as wide as it and seven languages high, in the box's colours, and snaps to whole languages; the chosen language is red and the one under the mouse teal. Its scroll bar lies over it.

2.2.10 Options on tall windows

lines 146–159

On windows at least 588 pixels high, the parchment behind the window, OK and Cancel in its corner, and the title on its ribbon.

2.2.11 Tree

lines 160–181

The tree's colours, found rows in amber, the rows' icons from the four atlases with an outline by kind and reach as `colour_key.csv` says, and greyed-out icons.

2.2.12 Tools

lines 182–190

Filters chips, no legend, Find first, and Clear and Expand all as two buttons that share the pane's width.

2.2.13 Find box

lines 191–208

Find's field in its 9-slice frame, 325 px wide, with the search picture while empty and the clearing cross while not; the hidden label and Include's hidden description.

2.2.14 Tooltip

lines 209–213

The tooltip's place, set by the script as numbers, 32 pixels from the pointer as both cursors are 32 pixels large, and its parchment look.

2.2.15 Civilisations

lines 214–229

The civilisations' icons from their atlas, and the tiles, six to a row, each with its icon and its name in Trajan Pro.

2.2.16 Scroll bars

lines 230–261

The game's scroll bars: the browser's are hidden, room is kept for the bar, and the bar is placed, sized and drawn from its arrows, body and knob, each from its sheet, as `fit()` sets them. The language list's bar has the combo box's arrows instead, centred on it, and its arrows and knob always show their first picture.

3 js/

1 file 41 lines

Where the scripts start: script.js, which index.html loads as a module. Every other script is in one of the four folders below it: js/model/, js/view/ and js/controller/ for the page, and js/tests/ for the tests.

Subchapters: 3.1 script.js

3.1 script.js

Folder: js/ 41 lines 1 function, 0 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/page.js, js/view/pictures.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/controller/updatehandler.js, and js/tests/run_test.js (only with `?test`)
Imported by: index.html

Where the page starts: its top-level code runs once, and `start()` starts the parts of the page in their order.

3.1.1 `Start-up` TOP-LEVEL CODE lines 24–41

Sets Simple UI from the saved choice first, so pictures load at once or never. Loads data.json and the visitor's language (all of them for `?test` or `?update`), then has `start()` start the page, starts its pictures, and lets the loading screen go; `?test` loads the tests and `?update` runs the updater. If loading fails, it shows the load error.

Called by *no function of the code*
Calls `start()`; in js/model/datahandler.js: `load()`, `useData()`, `loadLanguage()`, `loadLanguages()`, `preferredLanguage()`; in js/view/stringbuilder.js: `showLoadError()`, `showLoading()`; in js/view/pictures.js: `finishLoading()`, `startPictures()`; in js/controller/optionshandler.js: `savedSimple()`, `setSimple()`; in js/controller/updatehandler.js: `runUpdate()`; in js/tests/run_test.js: `loadFailed()`, `runTests()`

3.1.2 `start()` INTERNAL lines 12–22

Starts the page: gives the model the saved Include choices and the language, or the first language when the visitor's could not be loaded, starts the controllers of the tree and of the Options window, writes the fixed texts, has the page observe the model, draws it once with `redraw("start")` and reveals it. Runs once, when data.json and the language have loaded.

Called by `Start-up`
Calls in js/model/datahandler.js: `preferredLanguage()`, `recall()`; in js/model/model.js: `startModel()`; in js/view/stringbuilder.js: `writeFixed()`, `showPage()`; in js/view/page.js: `startPage()`, `redraw()`; in js/controller/nodetree.js: `startTree()`; in js/controller/optionshandler.js: `startOptions()`

4 js/model/

2 files 648 lines

The data and the tree's rules: datahandler.js looks after the data and the languages, and model.js holds the tree and its rules, and tells the page when they change. Neither touches the page.

Subchapters: 4.1 datahandler.js, 4.2 model.js

4.1 datahandler.js

Folder: js/model/ 351 lines 47 functions, 32 exported

Imports: nothing

Imported by: js/script.js, js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/optionsview.js, js/view/scrollbars.js, js/view/pictures.js, js/view/report.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/controller/updatehandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/static_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

Everything about data: loading, the chosen language and its text, sets of civilisations, saved choices, the address, and the updater.

4.1.1 `attempt(action)` EXPORTED lines 9–15

Runs a function and returns null instead of an error. Used where the browser may refuse, such as storage in a private window, so a refusal never breaks the page.

Called by `recall()`, `remember()`, `keepInAddress()`; in js/view/scrollbars.js: `hold()`, `drag()`; in js/view/pictures.js: `rooted()`, `pictureFiles()`; in js/tests/run_test.js: `restore()`; in js/tests/helpers_test.js: `tipLooks()`; in js/tests/interface_test.js: `optionTests()`; in js/tests/pictures_test.js: `simpleSwitch()`

Calls *no function of the code*

4.1.2 `recall(key)` EXPORTED line 17

Reads a saved choice from the browser's storage, or null. Remembers the visitor's language and options between visits.

Called by `preferredLanguage()`; in js/script.js: `start()`; in js/controller/optionshandler.js: `savedSimple()`, `startOptions()`; in js/tests/run_test.js: `prepare()`; in js/tests/helpers_test.js: `englishFirst()`, `tipLooks()`; in js/tests/interface_test.js: `optionTests()`; in js/tests/pictures_test.js: `simpleSwitch()`

Calls `attempt()`

4.1.3 `remember(key, value)` EXPORTED lines 19–21

Saves a choice in the browser's storage, or removes it when the value is null. Failures are ignored, as the page works without storage.

Called by in js/controller/optionshandler.js: `setInclude()`, `setRegional()`, `setSimple()`, `chooseView()`, `chooseTips()`, `choose()`; in js/tests/run_test.js: `restore()`; in js/tests/helpers_test.js: `englishFirst()`

Calls `attempt()`

4.1.4 `keepInAddress(key, value)` EXPORTED lines 23–27

Puts a choice into the address, as `?lng=` for the language, so that a shared link keeps it, without a new step in the browser's history. The only function that writes the address; failures are ignored.

Called by in js/controller/optionshandler.js: `choose()`

Calls `attempt()`

4.1.5 `load(url)` EXPORTED lines 29–37

Fetches a file as text and returns a promise. It uses XMLHttpRequest because, unlike `fetch()`, it also reads files when the page is opened from disk, as the tests in CI do.

Called by `loadLanguage()`, `fetchUpdate()`; in js/script.js: `Start-up`; in js/tests/run_test.js: `rounds()`; in js/tests/helpers_test.js: `edges()`; in js/tests/static_test.js: `scripts()`, `styles()`

Calls *no function of the code*

4.1.6 `fill(template, values)` EXPORTED lines 39–40

Replaces `{name}` placeholders in a text with values, leaving unknown ones alone. All text lives in the data with placeholders, so the code never builds sentences itself.

Called by `plural()`, `refresh()`; in js/view/stringbuilder.js: `chipName()`, `listFilters()`; in js/view/treeview.js: `count()`; in js/view/report.js: `showUpdating()`, `showUpdate()`, `showUpdateFailed()`; in js/tests/run_test.js: `reportLine()`, `runTests()`; in js/tests/behaviour_test.js: `fullName()`, `marking()`, `counts()`; in js/tests/interface_test.js: `languageTests()`

Calls *no function of the code*

4.1.7 `treeKeys(nodes, key)` EXPORTED lines 42–44

Lists every item (or group) name in a tree, top to bottom. Used to check the tree, to find groups that need text, and by the updater to know which items to look up.

Called by `missingText()`, `refresh()`; in js/tests/helpers_test.js: `keys()`

Calls *no function of the code*

4.1.8 `useLanguage(code)` EXPORTED lines 53–60

Switches the active language: its text, language tag, sorting, plural rules and number format. Everything that shows, sorts or counts text reads these, so one call changes them all.

Called by in js/model/model.js: `startModel()`, `chooseLanguage()`

Calls *no function of the code*

4.1.9 `nameOf(item)` EXPORTED line 62

Returns a tree item's or group's name in the current language, or its key if missing. Used wherever an item is shown or searched.

Called by in js/view/stringbuilder.js: `share()`, `chipName()`; in js/view/treeview.js: `label()`; in js/controller/nodetree.js: `search()`; in js/tests/helpers_test.js: `dropsGreyed()`, `fixtures()`; in js/tests/behaviour_test.js: `fullName()`, `branches()`, `lineTops()`, `standIns()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `languageTests()`, `findKey()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `findLooks()`; in js/tests/audit_test.js: `controlChecks()`

Calls *no function of the code*

4.1.10 `civName(civ)` EXPORTED line 64

Returns a civilisation's name in the current language, or its key if missing. Keeps the civilisation list translated.

Called by in js/view/stringbuilder.js: `listCivs()`; in js/tests/helpers_test.js: `expected()`

Calls *no function of the code*

4.1.11 `iconFile({ kind, picture })` INTERNAL line 66

Returns the path of a tree item's icon PNG from its kind and picture number, such as `assets/images/icons/nodes/unit/8.png`, for the updater's report of new icons.

Called by `addIcons()`

Calls *no function of the code*

4.1.12 `civFile(civ)` EXPORTED line 68

Returns the name of a civilisation's icon file: its name in lower case, unless `icons.names` in data.json gives another, such as `inca` for the Incas.

Called by in js/view/stringbuilder.js: `civIcon()`; in js/tests/pictures_test.js: `unpictured()`

Calls *no function of the code*

4.1.13 `byFile(a, b)` INTERNAL line 70

Sorts picture numbers or names as their PNG files sort by name, `1.png`, `10.png`, `100.png` and on to `11.png`, which is the order of the cells in an atlas.

Called by `cellOf()`, `addIcons()`

Calls *no function of the code*

4.1.14 `cellOf(names, name, file = String)` EXPORTED lines 74–79

Finds a picture's cell in its atlas, which holds the pictures in file-name order, row by row, with as many columns as the square root of their number, rounded up. The order is worked out once per list.

Called by in js/view/stringbuilder.js: `civIcon()`; in js/view/treeview.js: `dress()`

Calls `byFile()`

4.1.15 `plural(key, n)` EXPORTED lines 81–84

Picks the right form of a counted text, such as "1 civilisation" or "5 civilisations", by the language's plural rules, and fills in the formatted number.

Called by in js/view/stringbuilder.js: `listCivs()`; in js/view/treeview.js: `count()`, `highlight()`; in js/tests/helpers_test.js: `dropsGreyed()`; in js/tests/behaviour_test.js: `marking()`, `counts()`, `finding()`
Calls `fill()`

4.1.16 `plain(text)` EXPORTED lines 86–87

Lower-cases a text and strips accents, spaces and punctuation, so Find matches names however they are typed, in any language.

Called by in js/controller/nodetree.js: `search()`, `find()`, `redrawn()`; in js/tests/behaviour_test.js: `finding()`; in js/tests/interface_test.js: `findKey()`
Calls *no function of the code*

4.1.17 `localeFile(code)` EXPORTED line 89

Returns the path of a language's file, such as `locales/de.json`. Keeps the pattern in one place for loading and for the updater's downloads.

Called by `loadLanguage()`, `pack()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

4.1.18 `unpack(file)` EXPORTED lines 91–94

Turns data.json as stored into the shape used in memory, where overrides and text are kept per language, so other languages can be added beside English.

Called by `useData()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

4.1.19 `addLanguage(set, code, file)` EXPORTED lines 96–99

Adds one language file's overrides and text to the data in memory. Used when a language loads, and when a test reads back the files the updater would write.

Called by `loadLanguage()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

4.1.20 `useData(raw)` EXPORTED lines 101–105

Reads data.json and gives each civilisation one bit, so a set of civilisations is a single number that `&` and `|` combine quickly.

Called by in js/script.js: `Start-up`
Calls `unpack()`

4.1.21 `loadLanguage(code)` EXPORTED lines 107–109

Loads a language's file unless it is loaded already. A visitor downloads only the language shown, not all of them.

Called by `loadLanguages()`; in js/script.js: `Start-up`; in js/controller/optionshandler.js: `choose()`
Calls `load()`, `localeFile()`, `addLanguage()`

4.1.22 `loadLanguages()` EXPORTED line 111

Loads every language's file. The tests and the updater need them all.

Called by in js/script.js: `Start-up`
Calls `loadLanguage()`

4.1.23 `pack(set)` EXPORTED lines 113–124

Splits the data in memory back into files: data.json with English, and one file per other language. The updater writes these, and a test checks that nothing is lost.

Called by `fetchUpdate()`; in js/tests/static_test.js: `updaterTests()`
Calls `localeFile()`

4.1.24 `preferredLanguage()` EXPORTED lines 126–129

Chooses the language to start in: the one in the address (`?lng=`), else the visitor's saved choice, else English (Req. 10). Shared links and return visits keep their language; first visits are English.

Called by in js/script.js: `Start-up`, `start()`; in js/tests/helpers_test.js: `englishFirst()`
Calls `recall()`

4.1.25 `bit(civ)` INTERNAL lines 131–134

Returns a civilisation's bit, or stops with an error for an unknown name, so a typo in data.json cannot silently drop a civilisation.

Called by `setOf()`
Calls *no function of the code*

4.1.26 `size(set)` EXPORTED lines 136–140

Counts the civilisations in a set by counting its bits. Behind every number on the page.

Called by in js/model/model.js: `prune()`, `countOf()`; in js/tests/helpers_test.js: `fixtures()`; in js/tests/behaviour_test.js: `marking()`, `crowding()`, `beyond()`, `counts()`; in js/tests/interface_test.js: `optionTests()`; in js/tests/pictures_test.js: `simpleSwitch()`
Calls *no function of the code*

4.1.27 `setOf(rule)` EXPORTED lines 142–146

Turns an availability rule from data.json (`"all"`, a `has` list or a `lacks` list) into a set. Rules stay short in the file while the code works with sets.

Called by in js/model/model.js: `collect()`, `prune()`; in js/tests/static_test.js: `updaterTests()`
Calls `bit()`

4.1.28 `civsOf(set)` EXPORTED line 148

Turns a set back into a list of civilisations, in data.json order. Used to list the civilisations that fit.

Called by in js/model/model.js: `matchingCivs()`; in js/tests/helpers_test.js: `expected()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

4.1.29 `isScalar(value)` INTERNAL line 150

Tells whether a value is text, a number, true or false, or null. Helps `toJson()` decide what fits on one line.

Called by `isFlatList()`, `flat()`, `toJson()`
Calls *no function of the code*

4.1.30 `isFlatList(value)` INTERNAL line 152

Tells whether a value is a list of scalars only, which `toJson()` writes on one line.

Called by `toJson()`
Calls `isScalar()`

4.1.31 `flat(value)` INTERNAL lines 154–158

Writes a value as one line of JSON, with a space after each comma and colon. The building block of `toJson()`.

Called by `toJson()`
Calls `isScalar()`

4.1.32 `toJson(value, indent = "")` EXPORTED lines 160–177

Writes JSON the way data.json is laid out: short objects and lists on one line, the rest indented, and a list that ends an object of scalars opened on the object's line. Keeps the updater's files readable and their changes easy to review.

Called by `fetchUpdate()`; in js/tests/static_test.js: `updaterTests()`
Calls `isScalar()`, `isFlatList()`, `flat()`

4.1.33 `isLower(char)` INTERNAL line 179

Tells whether a character is a lower-case letter. `joinLines()` uses it to tell a hyphenated word from a real hyphen.

Called by `joinLines()`
Calls *no function of the code*

4.1.34 `capitalise(text)` INTERNAL

line 181

Upper-cases the first letter of a text, as some names in the game start in lower case.

Called by `fromExtra()`, `localName()`, `translate()`
Calls *no function of the code*

4.1.35 `fromExtra(name, extra)` EXPORTED

lines 183–187

Turns an entry of `extras` in data.json, an item aoe2techtree.net gives to no civilisation, into the node, owners and picture counts the updater gathers for every other item. A test checks the extras the same way.

Called by `addExtras()`; in js/tests/static_test.js: `badExtras()`
Calls `capitalise()`

4.1.36 `joinLines(text, settings)` EXPORTED

lines 189–200

Joins a game text broken with `<br>` into one line, by the language's rules for spaces and hyphens, and tidies spaces. The game breaks names to fit its buttons.

Called by `localName()`, `translate()`; in js/tests/static_test.js: `updaterTests()`
Calls `isLower()`

4.1.37 `localName(strings, node, settings, offset)` INTERNAL

lines 202–207

Finds an item's name in a language's game texts. If that name looks abbreviated, it takes the bold name from the item's help text instead, which is written out in full.

Called by `translate()`
Calls `joinLines()`, `capitalise()`

4.1.38 `availabilityOf(owners, civs)` EXPORTED

lines 209–214

Writes who has an item in its shortest form: `"all"`, a `has` list or a `lacks` list. Keeps data.json small; a test checks that every rule is stored this way.

Called by `refresh()`; in js/tests/static_test.js: `badExtras()`, `updaterTests()`
Calls *no function of the code*

4.1.39 `pictureOf(node, counts)` EXPORTED

lines 216–220

Picks an item's icon for data.json from aoe2techtree.net's data: its kind, the picture most civilisations show, and whether it is common, regional or unique, which the outlines and Include Regionals use.

Called by `refresh()`; in js/tests/static_test.js: `badExtras()`
Calls *no function of the code*

4.1.40 `missingText(set)` EXPORTED

lines 222–233

Lists, per language, the page texts and group names that are missing. A test and the updater's report use it, so no language shows a blank.

Called by `compare()`; in js/tests/static_test.js: `textTests()`
Calls `treeKeys()`

4.1.41 `gather(civs, trees)` INTERNAL

lines 235–253

Reads aoe2techtree.net's tree of every civilisation into three tables: each item's node, the civilisations that have it, and how often each of its pictures is used. The updater's first step.

Called by `refresh()`
Calls *no function of the code*

4.1.42 `addExtras(extras, { nodes, owners, pictures }, say)` INTERNAL

lines 255–263

Adds the items of `extras` in data.json to the gathered tables, and reports any that aoe2techtree.net now has itself, so they can be taken out of `extras`.

Called by `refresh()`
Calls `fromExtra()`

4.1.43 `translate(current, fresh, { keys, nodes, techtree, strings })` INTERNAL lines 265–276

Writes every civilisation's and item's name in every language from the game's texts, unless an override says otherwise, writing out abbreviated names from the help text.

Called by `refresh()`
Calls `capitalise()`, `joinLines()`, `localName()`

4.1.44 `compare(current, fresh, { keys, nodes, owners, say })` INTERNAL lines 278–297

Lists the changes for the updater's report: civilisations added or gone, items whose civilisations changed or that nobody has, renamed civilisations and items, items not yet in the tree, and missing texts.

Called by `refresh()`
Calls `missingText()`

4.1.45 `addIcons(fresh, say)` INTERNAL lines 299–307

Adds every new picture to the list of its atlas in `icons.atlas`, in file-name order, and reports it with its PNG, so the atlas can be built again.

Called by `refresh()`
Calls `byFile()`, `iconFile()`

4.1.46 `refresh(current, techtree, trees, strings)` INTERNAL lines 309–334

The updater's core: works out the civilisations, who has what, the icons and every name from aoe2techtree.net's data and the extras, dates the data, and writes the report. Stops if the tree names an unknown item, or one twice.

Called by `fetchUpdate()`
Calls `fill()`, `gather()`, `addExtras()`, `treeKeys()`, `availabilityOf()`, `pictureOf()`, `addIcons()`, `translate()`, `compare()`

4.1.47 `fetchUpdate()` EXPORTED lines 336–351

Downloads aoe2techtree.net's data, trees and texts, runs `refresh()`, and returns the report and the files that changed, data.json always. So a game update needs only a browser.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls `load()`, `refresh()`, `pack()`, `toJson()`

4.2 model.js

Folder: js/model/ 297 lines 53 functions, 25 exported

Imports: js/model/datahandler.js
Imported by: js/script.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/page.js, js/view/pictures.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/static_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

The tree and its rules: which items are shown, how marks pass on, what is greyed out, which civilisations fit and how many, and what Filters shows. It is the subject of the Observer pattern: only its own functions change it, and each then tells its observer, page.js, what changed. It never touches the page.

4.2.1 `subscribe(observer)` EXPORTED lines 11–13

Adds an observer, which `notify()` then calls after every change. Only page.js subscribes (Test 85).

Called by in js/view/page.js: `startPage()`
Calls *no function of the code*

4.2.2 `unsubscribe(observer)` EXPORTED lines 15–17

Takes an observer away again.

Called by *no function of the code*
Calls *no function of the code*

4.2.3 `notify(change)` INTERNAL line 19

Tells every observer what kind of change was made, once and as soon as `matching` is up to date: `"marks"`, `"tree"` or `"language"`. The observers read the rest from the model themselves. It is the model's own, so nothing else can announce a change.

Called by `setIncluded()`, `setRegionals()`, `chooseLanguage()`, `mark()`, `clear()`, `unmarkLine()`
Calls *no function of the code*

4.2.4 `setIncluded(on)` EXPORTED lines 21–25

Turns Include Unique and Universal on or off, builds the tree again, keeping marks, and tells the page that the tree changed.

Called by in js/controller/optionshandler.js: `setInclude()`
Calls `rebuild()`, `notify()`

4.2.5 `setRegionals(on)` EXPORTED lines 27–31

Turns Include Regionals on or off, builds the tree again, keeping marks, and tells the page that the tree changed.

Called by in js/controller/optionshandler.js: `setRegional()`
Calls `rebuild()`, `notify()`

4.2.6 `collect(nodes, parent)` INTERNAL lines 33–47

Turns the shown part of data.json's tree into the items the page uses, each with its key, a path that tells copies apart, its set of civilisations (all, for a group) and its links. Stops with an error for an item without availability.

Called by `rebuild()`
Calls in js/model/datahandler.js: `setOf()`

4.2.7 `whole(item)` EXPORTED line 49

The set of civilisations that have an item and everything under it, worked out once and kept on the item. A row's count of matching civilisations uses it.

Called by `rebuild()`; in js/tests/helpers_test.js: `fixtures()`
Calls *no function of the code*

4.2.8 `lift(node, kids)` INTERNAL lines 51–52

Hands the rows of a unique item that is left out to its place, noting the item in their path, so that without the Camel Scout the Camel Rider is a row of the Stable. Lent rows are dropped.

Called by `prune()`
Calls *no function of the code*

4.2.9 `prune(nodes, every = false, regionals = true)` INTERNAL lines 54–65

Leaves out what Options leaves out: the unique groups and unique items (keeping their rows), regional items without Include Regionals, and items that one, all or all but one civilisation have, unless something under them is shown.

Called by `rebuild()`
Calls `lift()`; in js/model/datahandler.js: `size()`, `setOf()`

4.2.10 `lent(kid)` INTERNAL line 67

Tells whether a row can be lent to an item that stands in for its own: neither unique nor marked `"own"`.

Called by `common()`
Calls *no function of the code*

4.2.11 `common(node)` INTERNAL line 69

Copies a row and the rows under it that can be lent, marked as copies, for an item that stands in for it.

Called by `borrowed()`
Calls `lent()`

4.2.12 `borrowed(nodes, key, line = false)` INTERNAL lines 71–78

Finds the item that a `"replaces"` names and returns the rows it lends; none if it counts as a unit or technology, as its rows are its own upgrades.

Called by `variants()`
Calls `common()`

4.2.13 `named(link)` INTERNAL line 80

Turns a `"replaces"` or `"excludes"` value, one name, a list or nothing, into a list.

Called by `variants()`, `pair()`, `siblings()`
Calls *no function of the code*

4.2.14 `variants(nodes)` INTERNAL lines 82–85

Adds under every item that names items it stands in for the rows those items lend, in the order named, so their technologies are written only once in data.json.

Called by `rebuild()`
Calls `named()`, `borrowed()`

4.2.15 `twin()` INTERNAL lines 87–94

Ties every lent row to its original, so both share one mark: marking Two-Man Saw under the Mule Cart marks it under the Lumber Camp too, with one entry in Filters.

Called by `rebuild()`
Calls *no function of the code*

4.2.16 `pair(link, list)` INTERNAL lines 96–103

Links items both ways by `"excludes"` or `"replaces"`, as rivals or as alternatives.

Called by `rebuild()`
Calls `named()`

4.2.17 `side(item, rival)` INTERNAL line 105

An item and everything under it, but its rival and what is under that, so the Stable's side leaves out the Xolotl Warrior inside it.

Called by `oppose()`
Calls *no function of the code*

4.2.18 `oppose()` INTERNAL lines 107–111

Gives every item the opponents that its must have greys out: everything on its rival's side, and the other way round.

Called by `rebuild()`
Calls `side()`

4.2.19 `siblings()` INTERNAL lines 113–119

Adds to the alternatives of an item that stands in for others the other items that stand in for the same, so marking the Folwark greys out the Settlement too.

Called by `rebuild()`
Calls `named()`

4.2.20 `rebuild()` INTERNAL lines 121–134

Builds the items afresh from data.json and the Include options, links copies, rivals and alternatives, keeps every mark by its path, and works out `matching` again (`fit()`). Runs at the start and whenever an Include option changes.

Called by `setIncluded()`, `setRegionals()`, `startModel()`
Calls `collect()`, `prune()`, `variants()`, `whole()`, `pair()`, `oppose()`, `siblings()`, `twin()`, `fit()`

4.2.21 `startModel(included, regionals, code)` EXPORTED lines 136–141

Takes the saved Include choices and the language, and builds the tree once, telling no one: the page is drawn afterwards, all at once, with `redraw("start")`. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls `rebuild()`; in js/model/datahandler.js: `useLanguage()`

4.2.22 `chooseLanguage(code)` EXPORTED lines 143–146

Makes a language the one the tree's names are in, and tells the page that the language changed.

Called by in js/controller/optionshandler.js: `setLanguage()`
Calls `notify()`; in js/model/datahandler.js: `useLanguage()`

4.2.23 `barred(item)` EXPORTED lines 148–149

Tells whether an item is greyed out: not marked must have itself, while one of its alternatives or opponents is (Req. 37).

Called by `locked()`; in js/view/treeview.js: `paint()`; in js/controller/nodetree.js: `toggle()`
Calls *no function of the code*

4.2.24 `within(item, top)` INTERNAL line 151

Tells whether an item lies anywhere under another.

Called by `locked()`
Calls *no function of the code*

4.2.25 `locked(item)` EXPORTED line 153

Tells whether a greyed-out item is also closed: unless its own rival is inside it, as the Xolotl Warrior is inside the Stable, which stays open so the Xolotl Warrior stays in sight.

Called by `behind()`; in js/view/treeview.js: `paint()`, `treeView()`; in js/controller/nodetree.js: `expand()`, `expandAll()`
Calls `barred()`, `within()`

4.2.26 `behind(item)` EXPORTED line 155

Tells whether an item lies inside a closed greyed-out item, where Find does not look.

Called by in js/controller/nodetree.js: `search()`; in js/tests/helpers_test.js: `dropsGreyed()`
Calls `locked()`

4.2.27 `setMark(item, mark, top = item)` INTERNAL lines 157–160

Marks an item and everything under it, groups only through their rows, skipping the rivals of the item marked. How a building or group is marked or cleared.

Called by `put()`, `clear()`, `unmarkLine()`
Calls *no function of the code*

4.2.28 `state(item, top = item)` EXPORTED lines 162–166

The mark an item shows: the one its own mark and all its rows share, or `"mixed"` when they differ, leaving out the rivals of the item asked about.

Called by `current()`, `boxState()`, `markOf()`, `filtersOf()`, `chipOf()`; in js/tests/behaviour_test.js: `branches()`, `rivals()`
Calls *no function of the code*

4.2.29 `isUnitOrTech(item)` EXPORTED lines 168–171

Tells whether an item is a unit or technology: an item with another item, or a `"lines"` group, above it. Buildings and groups mark everything in them; units and technologies pass marks along lines.

Called by `prerequisites()`, `current()`, `put()`, `reached()`, `boxState()`, `chipOf()`; in js/tests/behaviour_test.js: `standIns()`, `beyond()`, `lines()`
Calls *no function of the code*

4.2.30 `descendants(item)` INTERNAL line 173

Everything under an item, at every depth.

Called by `put()`, `reached()`, `boxState()`
Calls *no function of the code*

4.2.31 `covers(above, item)` INTERNAL line 175

Tells whether every civilisation that has an item also has the one above it, so that must have may pass up to it.

Called by `prerequisites()`
Calls *no function of the code*

4.2.32 `prerequisites(item)` INTERNAL lines 177–181

The units and technologies above an item that its must have passes up to, stopping at a building or at an item that some civilisations with it lack (Req. 34).

Called by `put()`, `mark()`
Calls `isUnitOrTech()`, `covers()`

4.2.33 `passedTo(item)` INTERNAL lines 183–187

The items a mark passes down to: along the item's line until it forks, and only those the same civilisations have (Req. 29, 30).

Called by `put()`, `reached()`
Calls *no function of the code*

4.2.34 `current(item)` INTERNAL line 189

The mark an item has now: its own for a unit or technology, otherwise what all of it shows.

Called by `reach()`, `mark()`
Calls `isUnitOrTech()`, `state()`

4.2.35 `put(item, kind)` INTERNAL lines 191–198

Sets or clears a mark: on everything in a building or group, or along a unit's or technology's line. Must lack also clears must have below that depended on it.

Called by `mark()`
Calls `isUnitOrTech()`, `setMark()`, `descendants()`, `prerequisites()`, `passedTo()`

4.2.36 `reached(item)` INTERNAL line 200

The rows a mark reached, which marking opens: down to the first fork, and the rows it passed to or everything inside.

Called by `reach()`
Calls `chainOf()`, `isUnitOrTech()`, `passedTo()`, `descendants()`

4.2.37 `reach(item, kind)` EXPORTED line 202

The rows a mark will reach, or none if it clears the item. The controller opens the way to them before it marks, so they come into sight before the greyed-out branches close (Req. 28, 31).

Called by in js/controller/nodetree.js: `toggle()`
Calls `current()`, `reached()`

4.2.38 `mark(item, kind)` EXPORTED lines 204–210

Marks an item must have or must lack, or clears it if it already has that mark, passes must have up to its prerequisites, works out `matching` again and tells the page that the marks changed.

Called by in js/controller/nodetree.js: `toggle()`
Calls `current()`, `put()`, `prerequisites()`, `fit()`, `notify()`

4.2.39 `clear()` EXPORTED lines 212–216

Clears every mark (Req. 17 to 19), works out `matching` again and tells the page that the marks changed.

Called by in js/controller/nodetree.js: `startTree()`
Calls `setMark()`, `fit()`, `notify()`

4.2.40 `boxState(item)` EXPORTED lines 218–221

What a row's box shows: a unit's or technology's own mark, or a dash if a row below has another; otherwise `state()`.

Called by in js/view/treeview.js: `paint()`
Calls `isUnitOrTech()`, `state()`, `descendants()`

4.2.41 `fit()` INTERNAL lines 223–227

Works out the civilisations that fit every mark, as `matching`, and returns the marked items.

Called by `rebuild()`, `mark()`, `clear()`, `unmarkLine()`
Calls *no function of the code*

4.2.42 `countOf(item)` EXPORTED lines 229–232

The two counts beside an item's name: the civilisations that fit the marks and have the item, or for a group anything in it, and those that fit and have the item or anything under it. The tree's view shows one or both.

Called by in js/view/treeview.js: `count()`
Calls in js/model/datahandler.js: `size()`

4.2.43 `matchingCivs()` EXPORTED line 234

The civilisations that fit every mark, for the list on the right.

Called by in js/view/stringbuilder.js: `listCivs()`
Calls in js/model/datahandler.js: `civsOf()`

4.2.44 `markOf(item)` EXPORTED line 236

The mark an item stands for in Filters: the one all of it shows, else its own.

Called by `filtersOf()`
Calls `state()`

4.2.45 `lineEnd(item)` EXPORTED lines 238–242

The last item of a line of upgrades that does not fork, or null if the item is a group or its line forks.

Called by `lineOf()`; in js/tests/helpers_test.js: `fixtures()`; in js/tests/behaviour_test.js: `branches()`, `lineTops()`, `counts()`; in js/tests/interface_test.js: `uniqueFilterTests()`
Calls *no function of the code*

4.2.46 `lineOf(item)` EXPORTED lines 244–251

The whole line of upgrades an item is on, from its top to its end, or just the item if it is on none. Filters names and clears lines with it.

Called by `alone()`, `filtersOf()`, `chipOf()`, `unmarkLine()`; in js/tests/helpers_test.js: `lines()`, `fixtures()`; in js/tests/behaviour_test.js: `branches()`, `lineTops()`, `lines()`; in js/tests/interface_test.js: `uniqueFilterTests()`
Calls `lineEnd()`

4.2.47 `chainOf(item)` EXPORTED lines 253–257

An item and the rows below it for as long as each has exactly one row under it. Marking opens these.

Called by `reached()`; in js/controller/nodetree.js: `expand()`
Calls *no function of the code*

4.2.48 `unique(item)` INTERNAL line 259

Tells whether data.json counts an item as unique.

Called by `alone()`
Calls *no function of the code*

4.2.49 `alone(item)` INTERNAL line 261

Tells whether an item is unique in a line that is not all unique, so it keeps its own entry in Filters (Req. 36).

Called by `held()`, `chipOf()`
Calls `unique()`, `lineOf()`

4.2.50 `held(item, mark)` INTERNAL

line 263

Tells whether a mark keeps an item's own entry in Filters: must have on an item that `alone()` picks.

Called by `filtersOf()`, `unmarkLine()`
Calls `alone()`

4.2.51 `filtersOf()` EXPORTED

lines 265–281

Works out the entries of Filters: one per marked building, group or line, but none for what a marked row above covers, and the unique items' own entries, in the tree's order.

Called by in js/view/stringbuilder.js: `listFilters()`
Calls `state()`, `held()`, `markOf()`, `lineOf()`

4.2.52 `chipOf(item, kind, own = false)` EXPORTED

lines 283–286

What an entry of Filters names: a line's deepest item with must have, unless the item keeps its own entry, and whether it reads "Full": a building or group marked in full, with more than one row (Req. 24). `chipName()` puts it into words.

Called by in js/view/stringbuilder.js: `chipName()`
Calls `lineOf()`, `alone()`, `isUnitOrTech()`, `state()`

4.2.53 `unmarkLine(item, kind, own = false)` EXPORTED

lines 288–297

Clears what an entry of Filters stands for: the marks of its kind along its line, or the item and everything in it, keeping the unique items' own entries, and tells the page that the marks changed.

Called by in js/controller/nodetree.js: `startTree()`
Calls `held()`, `lineOf()`, `setMark()`, `fit()`, `notify()`

5 js/view/

7 files 848 lines

What the page shows: stringbuilder.js writes its text, treeview.js draws the tree, optionsview.js the Options window, page.js holds them all as one view of the whole page, pictures.js holds what only the pictures need, scrollbars.js draws the game's scroll bars, and report.js shows the report window. No view imports a controller.

Subchapters: 5.1 stringbuilder.js, 5.2 treeview.js, 5.3 optionsview.js, 5.4 page.js, 5.5 scrollbars.js, 5.6 pictures.js, 5.7 report.js

5.1 stringbuilder.js

Folder: js/view/ 138 lines 17 functions, 3 objects, 14 exported

Imports: js/model/datahandler.js, js/model/model.js
Imported by: js/script.js, js/view/treeview.js, js/view/optionsview.js, js/view/page.js, js/view/scrollbars.js, js/view/pictures.js, js/view/report.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

The page's text: the fixed texts, the texts of the chosen language, the list of civilisations and Filters, the loading screen, and the small helpers every view uses. Its views of the texts, the civilisations and Filters are parts of the page in page.js.

5.1.1 `$(id)` EXPORTED line 6

Short for `document.getElementById`, used wherever the page is touched.

Called by `clone()`, `writeFixed()`, `writeTexts()`, `listCivs()`, `listFilters()`, `showPage()`, `showLoadError()`, `showLoading()`; in js/view/treeview.js: `replant()`, `highlight()`, `findText()`, `onChip()`, `wireTools()`, `startTreeView()`; in js/view/optionsview.js: `options()`, `setMenu()`, `buildLanguages()`, `showLanguage()`, `tick()`, `showView()`, `checked()`, `readers()`, `wireOptions()`, `onMenuKey()`, `wireLanguages()`; in js/view/pictures.js: `setView()`, `wireTip()`, `onPage()`, `finishLoading()`; in js/view/report.js: `openReport()`, `report()`, `showUpdate()`, `showUpdateFailed()`; in js/tests/run_test.js: `runTests()`; in js/tests/helpers_test.js: `picturesOn()`, `simpleSpot()`, `simpleTicked()`, `civLinks()`, `nameFor()`, `type()`, `pick()`, `reset()`, `beside()`, `dropsGreyed()`, `listed()`, `barProblems()`, `untipped()`, `tipGaps()`, `tipLooks()`; in js/tests/behaviour_test.js: `chipNames()`, `marking()`, `branches()`, `crowding()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuAudit()`, `menuKeys()`, `languageAudits()`, `optionKeys()`, `optionAudits()`, `filterFocus()`, `keyboard()`, `findKey()`, `reportTests()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `offParchment()`, `wrongSizes()`, `pictureLooks()`, `wrongBoxes()`, `unpictured()`, `findLooks()`, `toolLooks()`, `simpleSwitch()`, `barLooks()`; in js/tests/audit_test.js: `structureChecks()`, `controlChecks()`

Calls *no function of the code*

5.1.2 `clone(id)` EXPORTED line 8

Copies the element inside a template in index.html. Repeated markup lives in index.html (Rule 2); the code only fills it in.

Called by `civIcon()`, `listCivs()`, `listFilters()`, `showLoading()`; in js/view/treeview.js: `dress()`, `build()`; in js/view/optionsview.js: `buildLanguages()`; in js/view/scrollbars.js: `startBars()`; in js/view/report.js: `report()`, `showUpdate()`

Calls `$()`

5.1.3 `simple()` EXPORTED line 10

Tells whether Simple UI is on, from the `pictures` class on the page, which is what shows the pictures. One source of truth, readable from every script.

Called by `listCivs()`; in js/view/treeview.js: `dress()`, `wireTools()`; in js/view/optionsview.js: `setMenu()`; in js/view/scrollbars.js: `fit()`, `wire()`; in js/view/pictures.js: `wireTip()`, `finishLoading()`, `applySimple()`; in js/controller/optionshandler.js: `choices()`; in js/tests/run_test.js: `prepare()`; in js/tests/pictures_test.js: `strayBars()`

Calls *no function of the code*

5.1.4 `place(icon, { column, row, columns, rows })` EXPORTED lines 12–18

Tells an icon which cell of its atlas to show, as four numbers for CSS: its column and row, and the atlas's columns and rows. Scripts give CSS only numbers.

Called by `civIcon()`; in js/view/treeview.js: `dress()`
Calls *no function of the code*

5.1.5 `compose(node, template, values)` EXPORTED lines 20–25

Writes a text with placeholders into an element, putting elements such as the source link where their placeholder is. Lets a translated sentence contain a link.

Called by `writeFixed()`, `writeTexts()`; in js/view/treeview.js: `count()`
Calls *no function of the code*

5.1.6 `demand(actions, names)` EXPORTED lines 27–31

Checks that a controller has every action a view needs, and stops with an error that names the missing ones. treeview.js and optionsview.js call it as they start, so a missing action shows at once.

Called by in js/view/treeview.js: `startTreeView()`; in js/view/optionsview.js: `startOptionsView()`
Calls *no function of the code*

5.1.7 `writeFixed()` EXPORTED lines 33–42

Writes the texts that never change: the report window's English words, the source's, Microsoft's and the project's links, and Microsoft's notice around its links. Runs once at the start.

Called by in js/script.js: `start()`
Calls `$()`, `compose()`

5.1.8 `writeTexts()` INTERNAL lines 44–52

Writes every text of the chosen language: the page's `lang`, each element marked `data-text`, and the footer's source and date (Req. 5, 11).

Called by `textsView()`
Calls `compose()`, `$()`

5.1.9 `textsView` OBJECT, EXPORTED lines 54–58

The view of the texts: writes them again at the start and when the language changes.

Called by in js/view/page.js: `the top-level code`
Calls `writeTexts()`

5.1.10 `share()` INTERNAL lines 62–66

Notes the names that more than one row uses, so that Filters can say which one is meant.

Called by `filtersView()`
Calls in js/model/datahandler.js: `nameOf()`

5.1.11 `chipName(item, kind, own = false)` EXPORTED lines 68–74

Names an entry of Filters from what `chipOf()` gives: a shared name "in" its parent, and "Full" before a building or group marked in full (Req. 24).

Called by `listFilters()`; in js/tests/behaviour_test.js: `marking()`, `rivals()`
Calls in js/model/datahandler.js: `nameOf()`, `fill()`; in js/model/model.js: `chipOf()`

5.1.12 `civIcon(civ)` INTERNAL line 76

Makes a civilisation's icon from its template, showing its cell of the civilisations' atlas. Only used without Simple UI.

Called by `listCivs()`
Calls `place()`, `clone()`; in js/model/datahandler.js: `cellOf()`, `civFile()`

5.1.13 `listCivs()` INTERNAL lines 78–93

Lists the civilisations that fit the marks (`matchingCivs()`), translated and sorted (Req. 4), with icons and links to aoe2techtree.net without Simple UI; writes the count, and the hint when the list is empty.

Called by `civsView()`
Calls `$()`, `clone()`, `simple()`, `civIcon()`; in js/model/datahandler.js: `civName()`, `plural()`; in js/model/model.js: `matchingCivs()`

5.1.14 `civsView` OBJECT, EXPORTED line 95

The view of the civilisations: lists them again after every change.

Called by in js/view/page.js: `the top-level code`
Calls `listCivs()`

5.1.15 `listFilters()` INTERNAL lines 97–111

Makes one Filters button per entry (Req. 22, 23), with its mark and name, and a label telling screen readers what a click clears. Hides Filters when nothing is marked.

Called by `filtersView()`
Calls `$()`, `clone()`, `chipName()`; in js/model/datahandler.js: `fill()`; in js/model/model.js: `filtersOf()`

5.1.16 `filtersView` OBJECT, EXPORTED lines 113–118

The view of Filters: notes the shared names again when the tree or the language changed, then makes the buttons again.

Called by in js/view/page.js: `the top-level code`
Calls `share()`, `listFilters()`

5.1.17 `showPage()` EXPORTED lines 120–123

Shows the header and the content, which stay hidden until the page has started.

Called by in js/script.js: `start()`
Calls `$()`

5.1.18 `showLoadError(shown = true)` EXPORTED lines 125–127

Shows, or hides, the message that the data or a language could not be loaded.

Called by in js/script.js: `Start-up`; in js/controller/optionshandler.js: `choose()`
Calls `$()`

5.1.19 `spin(img)` INTERNAL line 131

Gives the loading picture its address, kept in `data-src` so that it loads only when the loading screen shows it.

Called by `showLoading()`
Calls *no function of the code*

5.1.20 `showLoading(shown = true)` EXPORTED lines 133–138

Shows or hides the loading screen, with its turning picture, made once, and makes the page behind it unreachable while it shows.

Called by in js/script.js: `Start-up`; in js/view/pictures.js: `finishLoading()`, `applySimple()`
Calls `$()`, `spin()`, `clone()`

5.2 treeview.js

Folder: js/view/ 260 lines 27 functions, 1 object, 11 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js
Imported by: js/view/page.js, js/view/pictures.js, js/controller/nodetree.js, js/tests/helpers_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

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The tree on the left as the visitor sees it: its rows with their names, boxes, icons and counts, which branches are open, what Find found, and the focus. It is a part of the page in page.js, and hands every click, touch and key to the actions of nodetree.js, its controller (Strategy).

5.2.1 `rowOf(item)` EXPORTED line 13

The elements of an item's row: its entry, row, name, "must lack" words, counts, arrow, list and icon. They are kept here, by item, so that the model's items hold no element (Test 85).

Called by in js/tests/helpers_test.js: `nodeOf()`, `twistyOf()`, `labelOf()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `counts()`, `tree()`; in js/tests/audit_test.js: `controlChecks()`
Calls *no function of the code*

5.2.2 `opened(row)` INTERNAL line 15

Tells whether a row's branch is open, read from its arrow's `aria-expanded`, which is the one place this is kept.

Called by `isOpen()`, `replant()`
Calls *no function of the code*

5.2.3 `isOpen(item)` EXPORTED line 17

Tells whether an item's branch is open.

Called by `shown()`, `count()`; in js/controller/nodetree.js: `branch()`, `key()`; in js/tests/behaviour_test.js: `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`; in js/tests/interface_test.js: `keyboard()`, `headingTests()`
Calls `opened()`

5.2.4 `shown(item)` EXPORTED line 19

Tells whether an item's row can be seen, that is, whether every branch above it is open.

Called by in js/controller/nodetree.js: `key()`; in js/tests/behaviour_test.js: `branches()`, `rivals()`, `lines()`, `finding()`; in js/tests/interface_test.js: `headingTests()`
Calls `isOpen()`

5.2.5 `heading(item)` INTERNAL line 21

Tells whether a group is only a heading inside an item, which has no box and cannot be marked.

Called by `focusOf()`, `itemAt()`, `build()`, `paint()`, `onClick()`
Calls *no function of the code*

5.2.6 `focusOf(item)` INTERNAL line 23

The part of a row that takes focus: a heading's arrow, or any other row's name.

Called by `focus()`
Calls `heading()`

5.2.7 `itemAt(target)` EXPORTED lines 25–28

Returns the tree item an element belongs to, but not a heading, so one listener on the tree serves every row, and the tooltip knows when it is over the tree.

Called by `onPress()`, `onRelease()`, `onMenu()`, `onClick()`; in js/view/pictures.js: `tipFor()`
Calls `heading()`

5.2.8 `dress(item)` INTERNAL lines 30–37

Gives a tree row its icon between its box and name, from its cell of its atlas, with its kind and reach for the outline, or takes it away with Simple UI on (Rules 9, 22).

Called by `replant()`, `treeView()`
Calls in js/model/datahandler.js: `cellOf()`; in js/view/stringbuilder.js: `simple()`, `place()`, `clone()`

5.2.9 `build(list, ul)` INTERNAL lines 39–73

Makes a row for each item from the template, with its arrow, and ids that tie the arrow and the list to the name for screen readers. A heading loses its box. Calls itself for the items below.

Called by `replant()`
Calls `heading()`; in js/view/stringbuilder.js: `clone()`

5.2.10 `endPress()` INTERNAL lines 75–81

Cancels a long press in progress and its highlight, when the finger moves or lifts, or the tree is drawn again, and notes when a touch ended.

Called by `replant()`, `onPress()`, `onRelease()`, `startTreeView()`
Calls *no function of the code*

5.2.11 `replant()` INTERNAL lines 83–91

Draws the tree from the model again, keeping the branches that were open open by their paths, and dresses each row. Runs at the start and whenever the tree is built again.

Called by `treeView()`
Calls `opened()`, `endPress()`, `build()`, `dress()`, `setOpen()`; in js/view/stringbuilder.js: `$()`

5.2.12 `label()` INTERNAL lines 93–95

Writes every row's name in the current language.

Called by `treeView()`
Calls in js/model/datahandler.js: `nameOf()`

5.2.13 `count(item)` INTERNAL lines 97–112

Writes an item's count of civilisations that fit the marks, from the model's `countOf()`: its own number, or "own–all" when it is closed with items under it, and a spoken version for screen readers.

Called by `setOpen()`, `treeView()`
Calls `isOpen()`; in js/model/datahandler.js: `plural()`, `fill()`; in js/model/model.js: `countOf()`; in js/view/stringbuilder.js: `compose()`

5.2.14 `setOpen(item, open)` EXPORTED lines 114–120

Opens or closes an item's branch, and rewrites its count, since a closed item shows two numbers.

Called by `replant()`, `treeView()`; in js/controller/nodetree.js: `expand()`, `branch()`, `toggle()`, `find()`, `enter()`, `key()`, `expandAll()`, `collapseAll()`; in js/tests/behaviour_test.js: `branches()`, `rivals()`, `standIns()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `toolLooks()`
Calls `count()`

5.2.15 `paint()` INTERNAL lines 122–136

Updates every box: its state for the eye, `aria-checked` for screen readers, the hidden "must lack" words, the bold of a row's own mark, and greyed-out rows and arrows.

Called by `treeView()`
Calls `heading()`; in js/model/model.js: `boxState()`, `barred()`, `locked()`

5.2.16 `treeView` OBJECT, EXPORTED lines 138–148

The view of the tree: draws it again when the tree was built again, dresses its rows when Simple UI changes, names them when the language changes, then closes greyed-out items, paints the boxes and writes every count. Then it hands the change to its controller's `redrawn()`, so that Find follows the tree as it is drawn now.

Called by in js/view/page.js: `the top-level code`
Calls `replant()`, `dress()`, `label()`, `setOpen()`, `paint()`, `count()`; in js/model/model.js: `locked()`

5.2.17 `focus(item, onTwisty = false)` EXPORTED lines 150–153

Moves focus to an item's row, or to its arrow when asked and it has one.

Called by in js/controller/nodetree.js: `enter()`, `key()`
Calls `focusOf()`

5.2.18 `highlight(found, searched)` EXPORTED lines 155–158

Highlights the items Find found, and announces how many there are, or that there are none (Req. 20, 21).

Called by in js/controller/nodetree.js: `search()`
Calls in js/model/datahandler.js: `plural()`; in js/view/stringbuilder.js: `$()`

5.2.19 `reveal(item)` EXPORTED lines 160–166

Scrolls the tree's own list just enough to show a row, and nothing around it, so the Find box being typed in never moves (Rule 18).

Called by in js/controller/nodetree.js: `find()`
Calls *no function of the code*

5.2.20 `findText()` EXPORTED line 168

What is typed in the Find box.

Called by `wireTools()`; in js/controller/nodetree.js: `redrawn()`
Calls in js/view/stringbuilder.js: `$()`

5.2.21 `onKey(e)` INTERNAL lines 170–176

Hands a key in the tree to the controller's `key()`, with Shift, the other modifier keys and whether an arrow has focus, and stops the browser's own action when the controller used the key (WCAG 2.1.1).

Called by `startTreeView()`
Calls *no function of the code*

5.2.22 `onPress(e)` INTERNAL lines 178–189

Starts a right-click, or a touch that becomes a long press after half a second, which marks must lack (Req. 3). The row shows as pressed when the long press is ready.

Called by `startTreeView()`
Calls `endPress()`, `itemAt()`

5.2.23 `onRelease(e)` INTERNAL lines 191–198

Ends a press: reports a hold on the item to the controller if the press was ready and ends on the same item, and swallows the click that follows.

Called by `startTreeView()`
Calls `endPress()`, `itemAt()`

5.2.24 `onMenu(e)` INTERNAL lines 200–206

Stops the browser's own menu over the tree, and reports the menu key to the controller. Right-clicks and long presses are left to `onPress()` and `onRelease()`.

Called by `startTreeView()`
Calls `itemAt()`

5.2.25 `onClick(e)` INTERNAL lines 208–214

Opens or closes a branch from its arrow, or a heading from its name, or reports a click on an item to the controller, unless the click ends a long press. What a gesture marks is the controller's to decide.

Called by `startTreeView()`
Calls `heading()`, `itemAt()`

5.2.26 `onChip(e)` INTERNAL lines 216–223

Has the controller clear the marks behind a Filters button, and moves focus to the next button, or to the heading after the last one (WCAG 2.4.3).

Called by `wireTools()`
Calls in js/view/stringbuilder.js: `$()`

5.2.27 `wireTools()` INTERNAL lines 225–245

Connects Expand all and Collapse all, which hand focus to each other without Simple UI, Clear (Req. 17 to 19), Find with its clearing cross and Enter, and the Filters buttons, each to its action. Runs once at the start.

Called by `startTreeView()`
Calls `findText()`, `onChip()`; in js/view/stringbuilder.js: `$()`, `simple()`

5.2.28 `startTreeView(controller)` EXPORTED

lines 247–260

Starts the tree's view with its controller's actions, checked with `demand()`, and connects keys, presses, moves that cancel a long press, the menu and clicks, then the tools around the tree. Runs once, from `startTree()`.

Called by in js/controller/nodetree.js: `startTree()`

Calls `onKey()`, `onPress()`, `endPress()`, `onRelease()`, `onMenu()`, `onClick()`, `wireTools()`; in js/view/stringbuilder.js: `$()`, `demand()`

5.3 optionsview.js

Folder: js/view/ 126 lines 16 functions, 2 objects, 9 exported

Imports: js/model/datahandler.js, js/view/stringbuilder.js
Imported by: js/view/page.js, js/view/pictures.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

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The Options window as the visitor sees it: the language menu, the boxes and Civ view, as they are ticked and picked before OK. It is a part of the page in page.js, and hands opening the window and OK to the actions of optionshandler.js, its controller (Strategy).

5.3.1 `options()` EXPORTED line 8

Lists the language menu's buttons. Used by the menu's keys and by the tests.

Called by `showLanguage()`, `picked()`, `onMenuKey()`; in js/tests/helpers_test.js: `pick()`; in js/tests/interface_test.js: `menuTests()`, `optionTests()`, `menuKeys()`; in js/tests/pictures_test.js: `barLooks()`; in js/tests/audit_test.js: `structureChecks()`
Calls in js/view/stringbuilder.js: `$()`

5.3.2 `setMenu(open, focus)` EXPORTED lines 10–15

Opens or closes the language menu, and moves focus if asked. With Simple UI it hides the settings while the list is open, so that the window keeps its size (Rule 18); without Simple UI the list lies over them, as the game's combo box does.

Called by `wireOptions()`, `onMenuKey()`, `wireLanguages()`; in js/tests/helpers_test.js: `reset()`, `tipGaps()`; in js/tests/pictures_test.js: `barLooks()`
Calls in js/view/stringbuilder.js: `$()`, `simple()`

5.3.3 `buildLanguages()` EXPORTED lines 17–26

Makes one button per language, named and tagged in its own language, so screen readers say each name right (WCAG 3.1.2). Runs once at the start.

Called by in js/controller/optionshandler.js: `startOptions()`
Calls in js/view/stringbuilder.js: `$()`, `clone()`

5.3.4 `showLanguage(code)` INTERNAL lines 28–31

Shows a language as chosen in the window: its name on the menu's button and its mark in the list. OK makes it the page's language.

Called by `the top-level code`, `optionsView()`, `wireLanguages()`
Calls `options()`; in js/view/stringbuilder.js: `$()`

5.3.5 `tick(id, on)` EXPORTED lines 33–36

Shows an Options box as ticked or not, for the eye and for screen readers.

Called by `toggle()`, `show()`; in js/view/pictures.js: `setTips()`; in js/controller/optionshandler.js: `setInclude()`, `setRegional()`, `setSimple()`, `startOptions()`
Calls in js/view/stringbuilder.js: `$()`

5.3.6 `showView(choice)` EXPORTED lines 38–40

Marks the chosen view among the Civ view radio buttons.

Called by `the top-level code`; in js/view/pictures.js: `setView()`
Calls in js/view/stringbuilder.js: `$()`

5.3.7 `picked()` INTERNAL line 42

The language marked in the window's list, which OK makes the page's language.

Called by `readers()`
Calls `options()`

5.3.8 `checked(id)` INTERNAL line 44

Tells whether an Options box is ticked.

Called by `toggle()`, `shown()`
Calls in js/view/stringbuilder.js: `$()`

5.3.9 `toggle(id)` INTERNAL line 46

Ticks or unticks an Options box in the window only; OK applies it.

Called by `wireOptions()`
Calls `tick()`, `checked()`

5.3.10 `readers` OBJECT, INTERNAL line 48

How the window reads the two settings that are no boxes: the language and Civ view.

Called by `shown()`
Calls `picked()`; in js/view/stringbuilder.js: `$()`

5.3.11 `shown(id)` EXPORTED line 52

What the window shows for a setting: its own reading, or its box.

Called by in js/controller/optionshandler.js: `apply()`
Calls `readers()`, `checked()`

5.3.12 `show(id, value)` EXPORTED line 54

Shows a value of a setting in the window: the language, Civ view, or a box.

Called by in js/controller/optionshandler.js: `reset()`, `choose()`
Calls `tick()`

5.3.13 `optionsView` OBJECT, EXPORTED lines 56–60

The view of the Options window: marks the page's language in it at the start and when the language changes.

Called by in js/view/page.js: `the top-level code`
Calls `showLanguage()`

5.3.14 `wireOptions()` INTERNAL lines 62–77

Connects the cogwheel button, which has the controller show the page's values in the window before it opens, OK, which has the controller apply the window, and the boxes, by click and by Space. Runs once at the start.

Called by `startOptionsView()`
Calls `setMenu()`, `toggle()`; in js/view/stringbuilder.js: `$()`

5.3.15 `onMenuKey(e)` INTERNAL lines 79–98

Handles the language menu's keys: Down and Up open it at the current language and move through it, Home and End jump, and Escape closes it and returns focus (WCAG 2.1.1).

Called by `wireLanguages()`
Calls `options()`, `setMenu()`; in js/view/stringbuilder.js: `$()`

5.3.16 `onScrollbar(e)` INTERNAL lines 100–101

Tells whether a click landed on the language list's own scroll bar, which Chrome reports as a click, so that scrolling the list does not close it.

Called by `wireLanguages()`
Calls *no function of the code*

5.3.17 `wireLanguages()` INTERNAL lines 103–120

Connects the language menu: its button, a click on a language or anywhere else in the list (Req. 11, 12), its keys, and closing when focus or a click goes elsewhere. Runs once at the start.

Called by `startOptionsView()`
Calls `setMenu()`, `onScrollbar()`, `showLanguage()`, `onMenuKey()`; in js/view/stringbuilder.js: `$()`

5.3.18 `startOptionsView(controller)` EXPORTED lines 122–126

Starts the Options window's view with its controller's actions, checked with `demand()`, and connects the language menu and the window. Runs once, from `startOptions()`.

Called by in js/controller/optionshandler.js: `startOptions()`
Calls `wireLanguages()`, `wireOptions()`; in js/view/stringbuilder.js: `demand()`

5.4 page.js

Folder: js/view/ 14 lines 3 functions, 2 exported

Imports: js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/optionsview.js
Imported by: js/script.js, js/view/pictures.js

The whole page as one view (Composite), and the model's only observer (Observer). It holds the views of the texts, Options, the tree, the civilisations and Filters, and passes each change to all of them, always in that order. Its update, `draw()`, is only for the model's notices; `redraw()` draws the page for the start and for Simple UI.

5.4.1 `draw(change)` INTERNAL line 8

Passes a change to every view, in order: the model's `"marks"`, `"tree"` and `"language"`, `"start"` once at the start, and `"look"` after Simple UI changes.

Called by `the top-level code`, `redraw()`
Calls *no function of the code*

5.4.2 `redraw(reason)` EXPORTED line 12

Draws the page again without a change of the model: `"start"` once, from `start()`, and `"look"` when Simple UI changes, from `applySimple()`.

Called by in js/script.js: `start()`; in js/view/pictures.js: `applySimple()`
Calls `draw()`

5.4.3 `startPage()` EXPORTED line 14

Has the model tell the page of its changes. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls in js/model/model.js: `subscribe()`

5.5 scrollbars.js

Folder: js/view/ 121 lines 8 functions, 2 exported
Imports: js/model/datahandler.js, js/view/stringbuilder.js
Imported by: js/view/pictures.js, js/tests/pictures_test.js

The game's scroll bars, which replace the browser's without Simple UI. A bar is placed after each list that scrolls, so it does not scroll with it, and CSS draws it from the numbers `fit()` gives it.

5.5.1 `fit(bar, box)` EXPORTED lines 16–30

Gives a list's bar its place along the list's right edge, its height and the knob's position, as numbers for CSS, and hides the bar with Simple UI or when the list holds no more than it shows, and the knob when the bar is too short for it.

Called by `fitAll()`, `wire()`
Calls in js/view/stringbuilder.js: `simple()`

5.5.2 `fitAll()` INTERNAL line 32

Fits every bar again. Runs when a list or what is in it changes size, when anything is added or taken out, when Simple UI changes and when the window does.

Called by `the top-level code`, `startBars()`
Calls `fit()`

5.5.3 `watch(box)` INTERNAL line 36

Has the size observer watch a list and everything directly in it, so a bar is fitted again when they change size.

Called by `the top-level code`, `startBars()`
Calls *no function of the code*

5.5.4 `hold(part, step)` INTERNAL lines 43–64

Makes an arrow or the bar's body act on a press, and again every 50 milliseconds after 400 while held, keeping hold of the pointer so the press stays while it leaves.

Called by `wire()`
Calls in js/model/datahandler.js: `attempt()`

5.5.5 `page(box, knob, y, line)` INTERNAL lines 66–69

Scrolls a list by a page, less a line, towards a press above or below the knob, and not at all once the knob reaches the pointer.

Called by `wire()`
Calls *no function of the code*

5.5.6 `drag(knob, box)` INTERNAL lines 71–88

Lets the knob be dragged: the list scrolls as far as the knob moves, scaled from the body's room to the list's.

Called by `wire()`
Calls in js/model/datahandler.js: `attempt()`

5.5.7 `wire(bar, box)` INTERNAL lines 90–108

Connects a bar: the arrows scroll a line, the body pages, the knob drags and the wheel over the bar scrolls the list; the list's scrolling fits the bar again. A line is 40 pixels, but in the language list without Simple UI it is one language, and the wheel there, over the list or its bar, scrolls a language at a time, so that the list always shows whole languages.

Called by `startBars()`
Calls `hold()`, `page()`, `drag()`, `fit()`; in js/view/stringbuilder.js: `simple()`

5.5.8 `startBars()` EXPORTED lines 110–121

Puts a bar from its template after every list that scrolls, connects it, and fits the bars again whenever anything about them changes. Runs once, from `startPictures()`.

Called by in js/view/pictures.js: `startPictures()`
Calls `wire()`, `watch()`, `fitAll()`; in js/view/stringbuilder.js: `clone()`

5.6 pictures.js

Folder: js/view/ 141 lines 19 functions, 7 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/optionsview.js, js/view/page.js, js/view/scrollbars.js
Imported by: js/script.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js

What only the pictures need: Civ view, list or tiles, the tooltip, and the pictures and loading screen that come with leaving Simple UI. It also starts the scroll bars.

5.6.1 `setView(choice)` EXPORTED lines 11–15

Shows the civilisations as a list or as tiles, and marks the choice among the radio buttons. Tiles are pictures, so CSS shows them only without Simple UI.

Called by in js/controller/optionshandler.js: `chooseView()`; in js/tests/run_test.js: `restore()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `showView()`

5.6.2 `setTips(on)` EXPORTED lines 19–22

Switches the tooltips on or off, and ticks the Tooltips box.

Called by in js/controller/optionshandler.js: `chooseTips()`; in js/tests/run_test.js: `prepare()`, `restore()`
Calls in js/view/optionsview.js: `tick()`

5.6.3 `branchTip(target)` INTERNAL lines 24–28

The tooltip of a branch's arrow, or of the name of a group inside a building, which has no box and opens on a click: to collapse the branch while it is open, to expand it while it is closed, and none on a locked branch, whose arrow is disabled.

Called by `tipFor()`
Calls *no function of the code*

5.6.4 `tipFor(target)` INTERNAL lines 30–35

Returns the tooltip's text for what the mouse is over: the marking hint over the tree, the text that an element's `data-tip` names over every control, the hint of a branch's arrow (`branchTip()`), or the address of a civilisation's or the footer's link.

Called by `wireTip()`
Calls `branchTip()`; in js/view/treeview.js: `itemAt()`

5.6.5 `wireTip()` INTERNAL lines 37–64

Moves the tooltip with the mouse, with its corner at the cursor picture's corner and inside the window, and puts it into the Options window while that is open. After every click it works the tooltip out again for what is under the mouse (`elementFromPoint`), so that it follows a button that changes, such as Expand all turning into Collapse all. It shows only for a mouse, without Simple UI and with Tooltips on.

Called by `startPictures()`
Calls `tipFor()`; in js/view/stringbuilder.js: `$()`, `simple()`

5.6.6 `screenPictures(width = screen.width, height = screen.height, ratio = devicePixelRatio)` EXPORTED lines 66–71

Works out the pictures for the screen from its size in device pixels: the parchment (`fhd`, `qhd`, `uhd` or `ultrawide`, and `fhd` on a phone) and the size of every other picture, `sd` or `4k`.

Called by `applySimple()`; in js/tests/helpers_test.js: `misjudged()`
Calls *no function of the code*

5.6.7 `rulesIn(rules)` INTERNAL line 83

Lists every rule of a style sheet, with the rules inside media queries.

Called by `filesOf()`
Calls *no function of the code*

5.6.8 `linksIn(text, base)` INTERNAL line 85

Lists the picture and font addresses in a rule's text, made whole from the style sheet's own address.

Called by `filesOf()`
Calls *no function of the code*

5.6.9 `filesOf(sheet, use)` INTERNAL lines 87–88

Lists the addresses in a style sheet's rules whose selectors pass a test.

Called by `pictureFiles()`
Calls `rulesIn()`, `linksIn()`

5.6.10 `rooted(selector)` INTERNAL lines 90–91

Tells whether a selector already holds for `<html>` or `<body>`, as for the parchment and the cursors, whose pictures show at once.

Called by `applySimple()`
Calls in js/model/datahandler.js: `attempt()`

5.6.11 `fits(src)` INTERNAL lines 93–97

Tells whether a picture is the one for the screen: the parchment of `data-background`, and the size of `data-assets`. Pictures without a size always fit.

Called by `pictureFiles()`
Calls *no function of the code*

5.6.12 `pictureFiles(use = () => true)` EXPORTED lines 99–100

Lists the pictures and fonts in the style sheets that the screen uses, so the loading screen can wait for them all.

Called by `finishLoading()`, `applySimple()`
Calls `filesOf()`, `fits()`; in js/model/datahandler.js: `attempt()`

5.6.13 `copy(src)` INTERNAL line 104

Loads a picture once into a kept image, so that the browser has it before the page shows it.

Called by `finishLoading()`, `applySimple()`
Calls *no function of the code*

5.6.14 `onPage()` INTERNAL line 106

The addresses of the images already in the page, but for the loading screen's own.

Called by `finishLoading()`
Calls in js/view/stringbuilder.js: `$()`

5.6.15 `done(img)` INTERNAL line 108

Waits for a kept image to be decoded, or to fail, so that waiting never stops.

Called by `finishLoading()`
Calls *no function of the code*

5.6.16 `ready(pictures, fonts)` INTERNAL lines 110–111

Tells whether every picture and font has loaded already.

Called by `finishLoading()`
Calls *no function of the code*

5.6.17 `finishLoading()` EXPORTED lines 113–123

Hides the loading screen once every picture and font has loaded, at once if they have already. Without Simple UI only; a later call takes over from an earlier one.

Called by `applySimple()`; in js/script.js: `Start-up`
Calls `onPage()`, `pictureFiles()`, `copy()`, `ready()`, `done()`; in js/view/stringbuilder.js: `simple()`, `showLoading()`, `$()`

5.6.18 `applySimple(on)` EXPORTED lines 125–136

The page's part of switching Simple UI (Req. 32, 33): the first time it picks the screen's pictures and sets `--pixel`, the size of one screen pixel; it shows the loading screen when leaving Simple UI, adds or removes the `pictures` class, and has `redraw()` draw the page's views again with `"look"`.

Called by in js/controller/optionshandler.js: `setSimple()`
Calls `screenPictures()`, `pictureFiles()`, `rooted()`, `copy()`, `finishLoading()`; in js/view/stringbuilder.js: `simple()`, `showLoading()`; in js/view/page.js: `redraw()`

5.6.19 `startPictures()` EXPORTED

lines 138–141

Starts the tooltip and the scroll bars. Runs once, after `start()`.

Called by in js/script.js: `Start-up`
Calls `wireTip()`; in js/view/scrollbars.js: `startBars()`

5.7 report.js

Folder: js/view/ 48 lines 5 functions, 5 exported

Imports: js/model/datahandler.js, js/view/stringbuilder.js
Imported by: js/controller/updatehandler.js, js/tests/run_test.js, js/tests/interface_test.js

---

The report window of the updater and the tests. It only shows what they found.

5.7.1 `openReport(title, status)` EXPORTED lines 4–11

Opens the report window, used by the tests and the updater, with a title and a status line, and empties it.

Called by `showUpdating()`; in js/tests/run_test.js: `runTests()`; in js/tests/interface_test.js: `reportTests()`
Calls in js/view/stringbuilder.js: `$()`

5.7.2 `report(line)` EXPORTED lines 13–17

Adds a line to the report window.

Called by `showUpdate()`; in js/tests/run_test.js: `runTests()`; in js/tests/interface_test.js: `reportTests()`
Calls in js/view/stringbuilder.js: `clone()`, `$()`

5.7.3 `showUpdating()` EXPORTED lines 19–20

Opens the report window for the updater, saying that it is loading the source's data.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls `openReport()`; in js/model/datahandler.js: `fill()`

5.7.4 `showUpdate({ messages, fresh, files })` EXPORTED lines 22–44

Shows what the updater found: each change, the counts of civilisations, items and languages, and each changed file to download; or that the update is blocked, when the new data cannot be used.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls `report()`; in js/model/datahandler.js: `fill()`; in js/view/stringbuilder.js: `$()`, `clone()`

5.7.5 `showUpdateFailed(error)` EXPORTED lines 46–48

Shows why the update failed.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls in js/model/datahandler.js: `fill()`; in js/view/stringbuilder.js: `$()`

6 js/controller/

3 files 177 lines

What the visitor's clicks, touches and keys mean: nodetree.js for the tree, optionshandler.js for the Options window, and updatehandler.js for the updater. They change the model or ask a view to change, and never touch the page.

Subchapters: 6.1 nodetree.js, 6.2 optionshandler.js, 6.3 updatehandler.js

6.1 nodetree.js

Folder: js/controller/ 79 lines 12 functions, 1 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/treeview.js
Imported by: js/script.js

What the visitor's clicks, touches and keys in the tree mean: marking, opening and closing, the arrow keys, Find, the buttons above the tree and the buttons of Filters. These are the actions treeview.js is started with (Strategy). They change the model, or ask the tree's view to open a branch or move the focus, and never touch the page. The view hands them each change of the model once it has drawn it, so they never subscribe to the model.

6.1.1 `expand(item)` INTERNAL lines 9–11

Opens an item and each only child below it, so opening Militia shows its whole line in one step, unless the item is greyed out and closed. Used by the arrows and the Right key.

Called by `branch()`, `key()`
Calls in js/model/model.js: `locked()`, `chainOf()`; in js/view/treeview.js: `setOpen()`

6.1.2 `branch(item)` INTERNAL line 13

Opens or closes a branch, from its arrow or a heading's name.

Called by `startTree()`
Calls `expand()`; in js/view/treeview.js: `isOpen()`, `setOpen()`

6.1.3 `toggle(item, kind)` INTERNAL lines 15–21

Handles a mark from a gesture or a key, unless the item is greyed out: opens the way to every row the mark will reach (`reach()`, Req. 28, 31), then calls `mark()`.

Called by `gesture()`, `key()`
Calls in js/model/model.js: `barred()`, `reach()`, `mark()`; in js/view/treeview.js: `setOpen()`

6.1.4 `gesture(item, how)` INTERNAL line 23

Turns what the tree's view reports into a mark: a click marks must have (Req. 3), and a long press, a right-click and the menu key must lack, as `key()` decides for Space and X.

Called by `startTree()`
Calls `toggle()`

6.1.5 `search(text)` INTERNAL lines 25–28

Notes the items whose names hold Find's text, but those inside closed greyed-out items (`behind()`), and has the view highlight them and say how many there are (Req. 20, 21).

Called by `find()`, `redrawn()`
Calls in js/model/datahandler.js: `plain()`, `nameOf()`; in js/model/model.js: `behind()`; in js/view/treeview.js: `highlight()`

6.1.6 `find(typed)` INTERNAL lines 30–38

Runs Find: searches, opens the branches down to the matches and shows the first (Req. 20, 21).

Called by `redrawn()`, `startTree()`
Calls `search()`; in js/model/datahandler.js: `plain()`; in js/view/treeview.js: `setOpen()`, `reveal()`

6.1.7 `redrawn(change)` INTERNAL lines 40–43

Keeps Find up to date once the tree's view has drawn a change: after marks it only searches again, so that no branch opens or closes, and after a new tree, a new language or the start it runs the whole of Find again, since the names or the rows changed. Nothing changes for `"look"`.

Called by `startTree()`
Calls `search()`, `find()`; in js/model/datahandler.js: `plain()`; in js/view/treeview.js: `findText()`

6.1.8 `enter()` INTERNAL lines 45–49

Opens the branches down to the first item Find found and moves focus to it, on Enter in its box.

Called by `startTree()`
Calls in js/view/treeview.js: `setOpen()`, `focus()`

6.1.9 `key(item, typed)` INTERNAL lines 51–71

Handles a key in the tree: Space marks must have and X must lack (Req. 3); Up, Down, Home and End move; Right and Left open, enter, leave and close branches (WCAG 2.1.1). Tells the view whether it used the key.

Called by `startTree()`
Calls `toggle()`, `expand()`; in js/view/treeview.js: `shown()`, `isOpen()`, `setOpen()`, `focus()`

6.1.10 `expandAll()` INTERNAL line 73

Opens every branch but those of closed greyed-out items.

Called by `startTree()`
Calls in js/model/model.js: `locked()`; in js/view/treeview.js: `setOpen()`

6.1.11 `collapseAll()` INTERNAL line 75

Closes every branch.

Called by `startTree()`
Calls in js/view/treeview.js: `setOpen()`

6.1.12 `startTree()` EXPORTED lines 77–79

Starts the tree's view with these actions, and with the model's `clear()` and `unmarkLine()` for Clear and the Filters buttons. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls `gesture()`, `branch()`, `key()`, `find()`, `enter()`, `expandAll()`, `collapseAll()`, `redrawn()`; in js/model/model.js: `clear()`, `unmarkLine()`; in js/view/treeview.js: `startTreeView()`

6.2 optionshandler.js

Folder: js/controller/ 87 lines 11 functions, 1 object, 6 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/optionsview.js, js/view/pictures.js
Imported by: js/script.js, js/tests/run_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js

What the settings of the Options window do: the language, the two Include boxes, Simple UI, Civ view and Tooltips, each saved and applied on OK. Its `reset()` and `apply()` are the actions optionsview.js is started with (Strategy). The language in use is the model's alone: only the one still loading is kept here, as `pending`.

6.2.1 `setLanguage(code)` EXPORTED lines 9–11

Makes a language the page's through the model's `chooseLanguage()`, which tells the page. Runs when a language is chosen and its file is loaded.

Called by `choose()`; in js/tests/run_test.js: `restore()`
Calls in js/model/model.js: `chooseLanguage()`

6.2.2 `setInclude(on)` EXPORTED lines 13–17

Switches Include Unique and Universal (Req. 13 to 15): has the model build the tree again, keeping marks, saves the choice and ticks the box.

Called by `choices()`; in js/tests/run_test.js: `prepare()`, `restore()`; in js/tests/behaviour_test.js: `standIns()`, `beyond()`; in js/tests/interface_test.js: `optionTests()`, `optionKeys()`, `optionAudits()`, `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `unpictured()`
Calls in js/model/datahandler.js: `remember()`; in js/model/model.js: `setIncluded()`; in js/view/optionsview.js: `tick()`

6.2.3 `setRegional(on)` EXPORTED lines 19–23

Switches Include Regionals: has the model build the tree again, keeping marks, saves the choice and ticks the box.

Called by `choices()`; in js/tests/run_test.js: `prepare()`, `restore()`
Calls in js/model/datahandler.js: `remember()`; in js/model/model.js: `setRegionals()`; in js/view/optionsview.js: `tick()`

6.2.4 `savedSimple()` EXPORTED line 25

Reads the saved Simple UI choice: off unless it was saved as on, so a first visit shows the pictures.

Called by in js/script.js: `Start-up`
Calls in js/model/datahandler.js: `recall()`

6.2.5 `setSimple(on)` EXPORTED lines 27–31

Switches Simple UI (Req. 32, 33): saves the choice, ticks the box and has `applySimple()` change the page. Simple UI is only how the page looks, so it does not go through the model.

Called by `choices()`; in js/script.js: `Start-up`; in js/tests/run_test.js: `prepare()`, `restore()`; in js/tests/interface_test.js: `reportTests()`; in js/tests/pictures_test.js: `barLooks()`, `pictureTests()`
Calls in js/model/datahandler.js: `remember()`; in js/view/optionsview.js: `tick()`; in js/view/pictures.js: `applySimple()`

6.2.6 `chooseView(choice)` INTERNAL lines 33–36

Chooses Civ view, list or tiles: saves the choice and shows it.

Called by `choices()`, `startOptions()`
Calls in js/model/datahandler.js: `remember()`; in js/view/pictures.js: `setView()`

6.2.7 `chooseTips(on)` INTERNAL lines 38–41

Switches Tooltips on or off: saves the choice and shows it.

Called by `choices()`, `startOptions()`
Calls in js/model/datahandler.js: `remember()`; in js/view/pictures.js: `setTips()`

6.2.8 `choices` OBJECT, INTERNAL lines 43–50

Every setting of the Options window as an entry: how to read its value now and how to set it. `reset()` and `apply()` go through it.

Called by `apply()`, `reset()`
Calls `choose()`, `setInclude()`, `setRegional()`, `setSimple()`, `chooseView()`, `chooseTips()`; in js/view/stringbuilder.js: `simple()`

6.2.9 `apply()` INTERNAL lines 52–54

Applies every setting whose window value differs from the page's, when OK is pressed.

Called by `startOptions()`
Calls `choices()`; in js/view/optionsview.js: `shown()`

6.2.10 `reset()` INTERNAL lines 56–58

Shows the page's values again in the window, dropping what was changed, each time the window opens.

Called by `startOptions()`
Calls `choices()`; in js/view/optionsview.js: `show()`

6.2.11 `choose(code)` INTERNAL lines 60–78

Chooses a language: loads its file if needed, as `pending` so that only the last one chosen is used, shows it, saves the choice, and has `keepInAddress()` put it in the address (`?lng=`), so a shared link keeps it. If the file cannot be loaded, it shows the load error and puts the window back on the language in use.

Called by `choices()`
Calls `setLanguage()`; in js/model/datahandler.js: `loadLanguage()`, `remember()`, `keepInAddress()`; in js/view/stringbuilder.js: `showLoadError()`; in js/view/optionsview.js: `show()`

6.2.12 `startOptions()` EXPORTED lines 80–87

Starts the Options window: ticks the two Include boxes as the model has them, starts the window's view with `reset()` and `apply()`, builds the language menu, and restores Civ view and Tooltips. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls `reset()`, `apply()`, `chooseView()`, `chooseTips()`; in js/model/datahandler.js: `recall()`; in js/view/optionsview.js: `tick()`, `startOptionsView()`, `buildLanguages()`

6.3 updatehandler.js

Folder: js/controller/ 11 lines 1 function, 1 exported

Imports: js/model/datahandler.js, js/view/report.js
Imported by: js/script.js

Runs the updater, which `?update` opens.

6.3.1 `runUpdate()` EXPORTED lines 4–11

Runs the updater: shows that it is loading, has datahandler.js fetch the new data and work out what changed, and has report.js show it, or why it failed. It changes nothing by itself.

Called by in js/script.js: `Start-up`
Calls in js/model/datahandler.js: `fetchUpdate()`; in js/view/report.js: `showUpdating()`, `showUpdate()`, `showUpdateFailed()`

7 js/tests/

7 files 2325 lines

The tests, loaded only when the address ends in `?test`. run_test.js runs them and helpers_test.js holds the tools they share; the other five hold the tests, which TESTS.md describes one by one.

Subchapters: 7.1 run_test.js, 7.2 helpers_test.js, 7.3 static_test.js, 7.4 behaviour_test.js, 7.5 interface_test.js, 7.6 pictures_test.js, 7.7 audit_test.js

---

7.1 run_test.js

Folder: js/tests/ 108 lines 7 functions, 2 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/pictures.js, js/view/report.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/static_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js
Imported by: js/script.js (only with `?test`)

Runs the tests and shows the report. Loaded only when the address ends in `?test`.

7.1.1 `merge(where)` INTERNAL lines 19–28

Combines the results into one line per test, failed if any of its checks failed, with the first details and the window's size. A test may check in several states and languages.

Called by `runTests()`
Calls *no function of the code*

7.1.2 `reportLine(result, tools)` INTERNAL lines 30–31

Writes a test's report line, such as "Test 15 failed: ...", from its number, name and details.

Called by `runTests()`
Calls in js/model/datahandler.js: `fill()`

7.1.3 `loadFailed()` EXPORTED lines 33–35

Marks the tests as failed when the page could not load, so CI never waits for a result that cannot come.

Called by in js/script.js: `Start-up`
Calls *no function of the code*

7.1.4 `prepare()` INTERNAL lines 37–47

Notes the visitor's language, options and saved choices, then sets the tests' starting point: Include unticked, Include Regionals ticked, Simple UI and Tooltips on.

Called by `runTests()`
Calls in js/model/datahandler.js: `recall()`; in js/view/stringbuilder.js: `simple()`; in js/view/pictures.js: `setTips()`; in js/controller/optionshandler.js: `setInclude()`, `setRegional()`, `setSimple()`

7.1.5 `restore(before)` INTERNAL lines 49–59

Puts back the visitor's language, options, view, address and saved choices after the tests.

Called by `runTests()`
Calls in js/model/datahandler.js: `attempt()`, `remember()`; in js/view/pictures.js: `setView()`, `setTips()`; in js/controller/optionshandler.js: `setInclude()`, `setRegional()`, `setSimple()`, `setLanguage()`; in js/tests/helpers_test.js: `reset()`

7.1.6 `rounds()` INTERNAL lines 61–82

Runs every test in three rounds, functional, accessibility and maintainability, in the order TESTS.md lists them, picking sample items afresh where the tree was rebuilt.

Called by `runTests()`
Calls in js/model/datahandler.js: `load()`; in js/tests/helpers_test.js: `pick()`, `codes()`, `fixtures()`; in js/tests/static_test.js: `styles()`, `scripts()`, `textTests()`, `tagTests()`, `sourceTests()`, `dataTests()`, `updaterTests()`; in js/tests/behaviour_test.js: `behaviourTests()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `uniqueFilterTests()`, `headingTests()`, `keyboardTests()`, `accessibilityTests()`, `reportTests()`; in js/tests/pictures_test.js: `pictureTests()`

7.1.7 `runTests()` EXPORTED

lines 84–108

Runs the tests, numbers them in the order they first ran, checks the 100-test limit (Rule 17), marks the page passed or failed for CI, and shows the report, failures first. Its Close leaves the tests: it opens the page's own address, without `?test`.

Called by in js/script.js: `Start-up`

Calls `prepare()`, `rounds()`, `restore()`, `merge()`, `reportLine()`; in js/model/datahandler.js: `fill()`; in js/view/stringbuilder.js: `$()`; in js/view/report.js: `openReport()`, `report()`; in js/tests/helpers_test.js: `check()`

7.2 helpers_test.js

Folder: js/tests/ 459 lines 79 functions, 65 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/optionsview.js, js/view/pictures.js

Imported by: js/tests/run_test.js, js/tests/static_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

Small tools the test files share: recording results, finding and measuring elements, acting like a visitor, judging pictures and scroll bars, and picking sample items.

7.2.1 `check(name, ok, detail = "")` EXPORTED line 15

Records one test result. Every test reports this way, and each name is one line in the report.

Called by `expect()`; in js/tests/run_test.js: `runTests()`; in js/tests/static_test.js: `updaterTests()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `gestures()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuKeys()`, `optionKeys()`, `filterFocus()`, `keyboard()`, `findKey()`, `reportTests()`, `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `pictureTests()`; in js/tests/audit_test.js: `audit()`

Calls *no function of the code*

7.2.2 `expect(name, offenders)` EXPORTED line 17

Records a test that passes when its list of offenders is empty, naming the first few, so a failure points at the exact items.

Called by in js/tests/static_test.js: `sourceTests()`, `textTests()`, `tagTests()`, `treeTests()`, `textDataTests()`, `updaterTests()`; in js/tests/behaviour_test.js: `crowding()`, `counts()`, `tree()`; in js/tests/interface_test.js: `spacingTests()`; in js/tests/pictures_test.js: `pictureLooks()`, `pictureTests()`

Calls `check()`

7.2.3 `wait(ms)` EXPORTED line 19

Waits a number of milliseconds. Used to hold a long press.

Called by `loaded()`; in js/tests/behaviour_test.js: `gestures()`

Calls *no function of the code*

7.2.4 `settled(img)` INTERNAL lines 21–22

Waits until an image has loaded or failed.

Called by `loaded()`

Calls *no function of the code*

7.2.5 `loaded(images)` EXPORTED line 24

Waits until every image has loaded or failed, or ten seconds have passed, so the tests check pictures only once they are there.

Called by `frameShape()`

Calls `settled()`, `wait()`

7.2.6 `phone()` EXPORTED line 26

Tells whether the window uses the phone layout: narrower than 720 px and at least 640 px tall.

Called by `spans()`

Calls *no function of the code*

7.2.7 `width(selector)` INTERNAL line 28

Measures the width of the first element a selector finds.

Called by `spans()`

Calls *no function of the code*

7.2.8 `spans()` EXPORTED lines 30–32

Tells whether the footer is as wide as the panes, or hidden in the phone layout.

Called by in js/tests/interface_test.js: `languageTests()`; in js/tests/pictures_test.js: `pictureLooks()`

Calls `phone()`, `width()`

7.2.9 `need(ok, problem)` EXPORTED line 34

Returns a problem in a list if a condition fails, and an empty list if it holds, so a test can list exactly what went wrong.

Called by `barProblems()`; in js/tests/static_test.js: `mvcProblems()`; in js/tests/pictures_test.js: `wrongSizes()`, `pictureLooks()`, `unpictured()`, `toolLooks()`, `simpleSwitch()`, `strayBars()`, `barLooks()`
Calls *no function of the code*

7.2.10 `same(a, b)` EXPORTED line 36

Tells whether two lists are equal, item by item. Used to compare lists of civilisations.

Called by in js/tests/static_test.js: `updaterTests()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `optionKeys()`, `filterFocus()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `toolLooks()`, `simpleSwitch()`
Calls *no function of the code*

7.2.11 `codes()` EXPORTED line 38

Lists the language codes from data.json.

Called by `englishFirst()`; in js/tests/run_test.js: `rounds()`; in js/tests/static_test.js: `textTests()`, `tagTests()`, `textDataTests()`, `updaterTests()`; in js/tests/interface_test.js: `menuTests()`, `menuKeys()`, `languageAudits()`, `accessibilityTests()`; in js/tests/pictures_test.js: `toolLooks()`
Calls *no function of the code*

7.2.12 `nodeOf(item)` EXPORTED line 40

The row of an item's box and name, from the tree's view, which keeps the elements away from the items.

Called by `dropsGreyed()`, `tipGaps()`, `tipLooks()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `crowding()`, `gestures()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`, `counts()`, `finding()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuAudit()`, `languageAudits()`, `spacingTests()`, `optionKeys()`, `filterFocus()`, `keyboard()`, `findKey()`, `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `unpictured()`, `toolLooks()`, `simpleSwitch()`, `barLooks()`, `pictureTests()`
Calls in js/view/treeview.js: `rowOf()`

7.2.13 `twistyOf(item)` EXPORTED line 42

The arrow of an item's row, if it has one.

Called by `chainOf()`; in js/tests/behaviour_test.js: `rivals()`, `standIns()`, `lines()`, `tree()`; in js/tests/interface_test.js: `keyboard()`, `headingTests()`; in js/tests/pictures_test.js: `toolLooks()`; in js/tests/audit_test.js: `controlChecks()`
Calls in js/view/treeview.js: `rowOf()`

7.2.14 `labelOf(item)` EXPORTED line 44

The name in an item's row.

Called by in js/tests/interface_test.js: `languageTests()`, `headingTests()`; in js/tests/pictures_test.js: `unpictured()`
Calls in js/view/treeview.js: `rowOf()`

7.2.15 `englishFirst()` EXPORTED lines 46–54

Tells whether a first visit, with no saved choice and a German browser, starts in the language the address names, or else in English (Req. 10). Puts the saved choice back afterwards.

Called by in js/tests/interface_test.js: `menuTests()`
Calls `codes()`; in js/model/datahandler.js: `recall()`, `remember()`, `preferredLanguage()`

7.2.16 `ui()` EXPORTED line 56

Returns the page's texts in the current language.

Called by `tipLooks()`; in js/tests/interface_test.js: `menuTests()`; in js/tests/pictures_test.js: `simpleSwitch()`
Calls *no function of the code*

7.2.17 `box(el)` EXPORTED line 58

Short for an element's `getBoundingClientRect()`: its place and size on the screen.

Called by `barProblems()`, `untipped()`, `tipLooks()`; in js/tests/interface_test.js: `menuTests()`, `spacingTests()`, `reportTests()`; in js/tests/pictures_test.js: `cut()`, `offParchment()`, `findLooks()`, `toolLooks()`, `barLooks()`
Calls *no function of the code*

7.2.18 `keys()` EXPORTED line 60

Lists every item in the full tree.

Called by `regionalKeys()`; in js/tests/static_test.js: `sourceTests()`, `textTests()`, `dataTests()`, `updaterTests()`; in js/tests/interface_test.js: `optionTests()`
Calls in js/model/datahandler.js: `treeKeys()`

7.2.19 `describe(el)` EXPORTED lines 62–65

Names an element briefly, such as `button#clear` or a tree row's text, for failure messages.

Called by `pictures()`, `picturesOn()`, `barProblems()`, `untipped()`; in js/tests/interface_test.js: `spacingTests()`; in js/tests/pictures_test.js: `offParchment()`, `strayCursors()`, `wrongBoxes()`, `toolLooks()`, `strayBars()`, `ownBars()`; in js/tests/audit_test.js: `focusChecks()`, `structureChecks()`, `controlChecks()`, `colourChecks()`, `layoutChecks()`, `mediaChecks()`
Calls *no function of the code*

7.2.20 `rendered(el)` EXPORTED lines 67–68

Tells whether an element can be seen: laid out, not hidden, and not text for screen readers only. The accessibility checks look at what is rendered.

Called by `everything()`, `focusables()`, `visibleTexts()`, `untipped()`; in js/tests/interface_test.js: `optionTests()`; in js/tests/pictures_test.js: `offParchment()`, `findLooks()`, `toolLooks()`, `simpleSwitch()`, `strayBars()`; in js/tests/audit_test.js: `structureChecks()`, `controlChecks()`
Calls *no function of the code*

7.2.21 `everything()` EXPORTED line 70

Lists every element that can be seen on the page.

Called by `pictures()`; in js/tests/interface_test.js: `spacingTests()`; in js/tests/pictures_test.js: `strayCursors()`, `wrongSizes()`; in js/tests/audit_test.js: `audit()`
Calls `rendered()`

7.2.22 `styles(el)` EXPORTED line 72

Gets an element's styles and those of its `::before`, `::after` and `::marker` parts, so decorations are checked too.

Called by `pictured()`, `drawn()`; in js/tests/pictures_test.js: `cut()`, `wrongSizes()`; in js/tests/audit_test.js: `mediaChecks()`
Calls *no function of the code*

7.2.23 `pictured(el)` EXPORTED lines 74–76

Tells whether an element shows a picture through CSS: a background, list marker, border image, content or cursor, in itself or its parts.

Called by `pictures()`; in js/tests/audit_test.js: `mediaChecks()`
Calls `styles()`

7.2.24 `grade()` EXPORTED line 82

The size of pictures the page chose, `sd` or `4k`, from `data-assets`.

Called by `comboArrow()`, `barProblems()`; in js/tests/pictures_test.js: `cut()`, `strayCursors()`, `pictureLooks()`, `wrongBoxes()`, `unpictured()`
Calls *no function of the code*

7.2.25 `fitted(width = screen.width, height = screen.height, ratio = devicePixelRatio)` EXPORTED lines 84–89

Works out on its own, apart from the page's code, which parchment a screen should get from its size and scale, to judge `screenPictures()`.

Called by `misjudged()`; in js/tests/pictures_test.js: `pictureLooks()`
Calls *no function of the code*

7.2.26 `misjudged()` EXPORTED lines 96–99

Lists the screens, from phones to ultrawide and from 100% to 300%, for which `screenPictures()` picks another parchment or picture size than `fitted()`.

Called by in js/tests/pictures_test.js: `pictureLooks()`
Calls `fitted()`; in js/view/pictures.js: `screenPictures()`

7.2.27 `fitting(src, background)` EXPORTED lines 101–105

Tells whether a picture's address suits a parchment: its own parchment, or the `_sd` and `_x1` pictures with fhd and qhd and the `_4k` and `_x2` ones with uhd and ultrawide.

Called by in js/tests/pictures_test.js: `wrongSizes()`
Calls *no function of the code*

7.2.28 `files(value)` INTERNAL line 107

Reads the addresses out of a CSS value.

Called by `urlsOf()`
Calls *no function of the code*

7.2.29 `urlsOf(style)` EXPORTED lines 109–110

Lists the picture addresses in an element's style: background, border image, list marker, content and cursor.

Called by `frameShape()`, `sprite()`; in js/tests/pictures_test.js: `wrongSizes()`
Calls `files()`

7.2.30 `pictures()` EXPORTED lines 112–113

Lists every picture on the page: images and everything CSS draws with a picture, the page's own background included. Simple UI must leave it empty (Req. 33).

Called by `picturesOn()`; in js/tests/pictures_test.js: `simpleSwitch()`
Calls `everything()`, `pictured()`, `describe()`

7.2.31 `picturesOn()` EXPORTED lines 115–116

Tells whether the main pictures show: those of the page, of its body and of the Options button.

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls `describe()`, `pictures()`; in js/view/stringbuilder.js: `$()`

7.2.32 `rgb(colour)` INTERNAL line 118

Reads the numbers of a CSS colour.

Called by `yellow()`, `tone()`
Calls *no function of the code*

7.2.33 `yellow(colour)` EXPORTED lines 120–123

Tells whether a colour is yellow: red and green high and close, blue clearly lower (Req. 20).

Called by in js/tests/behaviour_test.js: `finding()`
Calls `rgb()`

7.2.34 `tone(colour)` INTERNAL lines 125–131

Names a colour as white, black, green, red or other, loosely enough for light and dark colours.

Called by `drawn()`
Calls `rgb()`

7.2.35 `drawn(box)` EXPORTED lines 133–141

Names the mark drawn on a checkbox, such as "green tick", "red cross" or "white dash", from the borders of its `::before` and `::after`, leaving out parts that draw nothing (Req. 14, 23, 26).

Called by in js/tests/behaviour_test.js: `branches()`; in js/tests/pictures_test.js: `wrongBoxes()`
Calls `styles()`, `tone()`

7.2.36 `simpleSpot()` EXPORTED lines 145–148

Returns where the Simple UI box is, to check that switching Simple UI does not move it (Rule 18).

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls in js/view/stringbuilder.js: `$()`

7.2.37 `simpleTicked(on)` EXPORTED lines 150–151

Tells whether the Simple UI box shows the given state, for the eye and for screen readers.

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls in js/view/stringbuilder.js: `$()`

7.2.38 `civLinks()` EXPORTED line 153

Counts the civilisations in the list that are links.

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls in js/view/stringbuilder.js: `$()`

7.2.39 `focusables()` EXPORTED lines 155–156

Lists every element that can be seen and reached with the keyboard.

Called by in js/tests/audit_test.js: `focusChecks()`, `controlChecks()`, `layoutChecks()`
Calls `rendered()`

7.2.40 `textOf(node)` INTERNAL lines 158–161

Reads an element's text as a screen reader would, leaving out parts marked `aria-hidden`.

Called by `nameFor()`
Calls *no function of the code*

7.2.41 `tidy(text)` EXPORTED line 163

Turns runs of spaces into single spaces and trims the ends.

Called by `nameFor()`; in js/tests/interface_test.js: `languageTests()`; in js/tests/audit_test.js: `layoutChecks()`
Calls *no function of the code*

7.2.42 `nameFor(el)` EXPORTED lines 165–171

Works out an element's accessible name much as a browser does: from `aria-labelledby`, `aria-label`, its label or its text. Lets the tests check that controls are named (WCAG 4.1.2).

Called by in js/tests/interface_test.js: `menuTests()`; in js/tests/audit_test.js: `controlChecks()`
Calls `tidy()`, `textOf()`; in js/view/stringbuilder.js: `$()`

7.2.43 `squash(text)` EXPORTED line 173

Lower-cases a text and keeps only letters and digits, for loose comparisons.

Called by in js/tests/audit_test.js: `controlChecks()`
Calls *no function of the code*

7.2.44 `visibleTexts(el)` EXPORTED lines 175–182

Lists the visible pieces of text inside an element. Used to find stray placeholders, and to compare names with what can be seen (WCAG 2.5.3).

Called by in js/tests/interface_test.js: `languageTests()`; in js/tests/audit_test.js: `controlChecks()`
Calls `rendered()`

7.2.45 `keyDown(target, key)` EXPORTED lines 184–186

Sends a key press to an element, as a visitor's keyboard would.

Called by `tipLooks()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`, `counts()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `menuKeys()`, `languageAudits()`, `optionKeys()`, `filterFocus()`, `keyboard()`, `findKey()`, `headingTests()`; in js/tests/pictures_test.js: `pictureTests()`
Calls *no function of the code*

7.2.46 `tap(el)` EXPORTED lines 188–192

Clicks an element with mouse pointer events, as a visitor would.

Called by `dropsGreyed()`, `tipGaps()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `crowding()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`, `counts()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuAudit()`, `languageAudits()`, `spacingTests()`, `optionKeys()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `toolLooks()`, `simpleSwitch()`, `barLooks()`, `pictureTests()`
Calls *no function of the code*

7.2.47 `type(text)` EXPORTED lines 194–197

Types a text into Find, as a visitor would.

Called by `reset()`, `dropsGreyed()`, `tipGaps()`; in js/tests/behaviour_test.js: `finding()`; in js/tests/interface_test.js: `findKey()`; in js/tests/pictures_test.js: `findLooks()`
Calls in js/view/stringbuilder.js: `$()`

7.2.48 `pick(code)` EXPORTED lines 199–204

Chooses a language through the Options window, its menu and OK, as a visitor would.

Called by in js/tests/run_test.js: `rounds()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `languageAudits()`; in js/tests/pictures_test.js: `toolLooks()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `options()`

7.2.49 `reset()` EXPORTED lines 206–213

Returns the page to its start: Find empty, no marks, every branch closed, the menu and the Options window closed, nothing focused. Tests start from here.

Called by `dropsGreyed()`, `tipGaps()`; in js/tests/run_test.js: `restore()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `crowding()`, `gestures()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuAudit()`, `menuKeys()`, `languageAudits()`, `spacingTests()`, `optionKeys()`, `optionAudits()`, `filterFocus()`, `keyboard()`, `findKey()`, `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `barLooks()`, `pictureTests()`
Calls `type()`; in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `setMenu()`

7.2.50 `beside()` EXPORTED lines 215–220

Tells whether Find's message is on one line, right of the box and level with it (Req. 21).

Called by in js/tests/behaviour_test.js: `finding()`; in js/tests/pictures_test.js: `findLooks()`
Calls in js/view/stringbuilder.js: `$()`

7.2.51 `dropsGreyed(rival)` EXPORTED lines 222–239

Tests that Find drops a match once a mark greys out its branch: marks a sample's rival, or that rival's own, until a branch is closed and greyed out, searches for an item in it without the mark, marks again, and checks that the match loses its yellow and that the count beside Find agrees (Test 31; Req. 20, 21).

Called by in js/tests/behaviour_test.js: `finding()`
Calls `reset()`, `tap()`, `nodeOf()`, `type()`; in js/model/datahandler.js: `nameOf()`, `plural()`; in js/model/model.js: `behind()`; in js/view/stringbuilder.js: `$()`

7.2.52 `listed()` EXPORTED line 241

Lists the civilisations the page shows.

Called by in js/tests/behaviour_test.js: `marking()`, `crowding()`, `lineTops()`, `parents()`, `rivals()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `optionKeys()`, `filterFocus()`
Calls in js/view/stringbuilder.js: `$()`

7.2.53 `expected(set)` EXPORTED line 243

Lists the names a set of civilisations should show, translated and sorted, to compare with `listed()`.

Called by in js/tests/behaviour_test.js: `marking()`, `lineTops()`, `parents()`, `rivals()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `filterFocus()`
Calls in js/model/datahandler.js: `civsOf()`, `civName()`

7.2.54 `below(item, top)` EXPORTED lines 245–248

Tells whether an item sits anywhere under another.

Called by in js/tests/behaviour_test.js: `branches()`, `parents()`, `standIns()`
Calls *no function of the code*

7.2.55 `itemAbove(item)` EXPORTED lines 250–253

Tells whether an item has another item above it, or sits in a group marked `"lines"`, which makes it a unit or technology. The tests work this out on their own, apart from the page's code.

Called by `onTop()`, `inner()`, `fixtures()`
Calls *no function of the code*

7.2.56 `chainOf(item)` EXPORTED lines 255–259

The rows that marking an item should open: the item and those below it while each has one row under it, those with an arrow only. Worked out apart from the page's own `chainOf()`.

Called by in js/tests/behaviour_test.js: `tree()`; in js/tests/interface_test.js: `keyboard()`
Calls `twistyOf()`

7.2.57 `hex(code)` INTERNAL line 263

Turns a colour written as six hex digits into the `rgb()` form the browser reports.

Called by `edges()`
Calls *no function of the code*

7.2.58 `edges()` EXPORTED lines 265–270

Reads `colour_key.csv` into the outline colour of each kind and reach of tree icon, so the tests compare the page with the file.

Called by in js/tests/pictures_test.js: `unpictured()`
Calls `hex()`; in js/model/datahandler.js: `load()`

7.2.59 `painted()` EXPORTED line 272

Waits until the browser has drawn the page once more, so that the scroll bars have been fitted.

Called by `tipGaps()`; in js/tests/pictures_test.js: `strayBars()`
Calls *no function of the code*

7.2.60 `frameShape(el, columns, rows)` INTERNAL lines 274–278

The height of one picture in a sheet divided by its width, from the sheet's own size, so a test can tell a picture drawn in proportion from a stretched one.

Called by `barProblems()`
Calls `urlsOf()`, `loaded()`

7.2.61 `sprite(el)` INTERNAL lines 280–284

Describes how an element draws its picture: the file, its size, its repeat and its position.

Called by `barProblems()`
Calls `urlsOf()`

7.2.62 `comboArrow()` INTERNAL line 286

The size of the combo box's arrow and how far it sits in from the box's right edge, in CSS pixels: 19 by 14 and 2 pixels in, or on a 4K screen 35 by 28 and 4 of the screen's own pixels.

Called by `barProblems()`
Calls `grade()`

7.2.63 `barProblems(bar, list)` EXPORTED lines 288–322

Lists what is wrong with a list's scroll bar: not along its right edge in the room kept for it, the wrong pictures or the wrong part of them, arrows, body and knob not equally wide, out of proportion, or the knob not where the list is scrolled to. The language list's bar has the combo box's arrows instead, centred on the bar and of the size `comboArrow()` gives.

Called by in js/tests/pictures_test.js: `strayBars()`
Calls `describe()`, `grade()`, `box()`, `comboArrow()`, `frameShape()`, `need()`, `sprite()`; in js/view/stringbuilder.js: `$()`

7.2.64 `hover(el, x, y)` EXPORTED lines 324–326

Moves the mouse over an element, as a visitor would.

Called by `untipped()`, `tipLooks()`
Calls *no function of the code*

7.2.65 `untipped(scope)` INTERNAL lines 330–335

Moves the mouse over every control and link in a part of the page that can be seen, is not disabled and is outside the report, and names those whose tooltip is missing or, but for Find's, longer than 100 characters, and the links whose tooltip is not their address.

Called by `tipGaps()`
Calls `rendered()`, `hover()`, `box()`, `describe()`; in js/view/stringbuilder.js: `$()`

7.2.66 `tipGaps()` EXPORTED lines 337–351

Lists the controls and links without a tooltip of at most 100 characters (`untipped()`): on the page with every branch open, a mark and a search, then in the Options window and with its language list open (Test 49).

Called by in js/tests/pictures_test.js: `pictureTests()`
Calls `tap()`, `nodeOf()`, `fixtures()`, `type()`, `painted()`, `untipped()`, `reset()`; in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `setMenu()`

7.2.67 `tipLooks()` EXPORTED lines 353–377

Tests the tooltip: over the tree at the cursor's corner, inside the window and in its parchment colour; over Include with Include's own tooltip, inside the Options window, while the description stays hidden but still describes the box to screen readers; and none once Tooltips is unticked, which is remembered, until Space ticks it again (Test 49).

Called by in js/tests/pictures_test.js: `pictureTests()`
Calls `box()`, `nodeOf()`, `hover()`, `ui()`, `keyDown()`; in js/model/datahandler.js: `recall()`, `attempt()`; in js/view/stringbuilder.js: `$()`

7.2.68 `shownKeys()` EXPORTED line 379

Lists the items currently in the tree, to compare the default tree with the full one.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

7.2.69 `uniqueKeys()` EXPORTED lines 381–388

Lists the items the default tree leaves out as unique: those inside unique groups and those aoe2techtree.net calls unique.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

7.2.70 `ownKeys()` EXPORTED lines 390–396

Lists the items marked `"own"` and everything under them, which items standing in for theirs never show.

Called by in js/tests/behaviour_test.js: `standIns()`
Calls *no function of the code*

7.2.71 `alternativesOf(key)` EXPORTED lines 398–406

Lists the items that marking an item must have should grey out, from data.json alone: the items it names, the items that name it, and those that name what it names (Req. 37).

Called by in js/tests/behaviour_test.js: `standIns()`
Calls *no function of the code*

7.2.72 `regionalKeys()` EXPORTED line 408

Lists the items aoe2techtree.net calls regional, which Include Regionals leaves out when unticked.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls `keys()`

7.2.73 `outside(nodes, regional)` EXPORTED lines 410–412

Lists the items of a tree outside regional items and all under them: what the tree shows with Include ticked and Include Regionals unticked.

Called by in js/tests/interface_test.js: `optionTests()`
Calls *no function of the code*

7.2.74 `keptPath()` EXPORTED lines 414–415

Finds the path of a leaf to mark while an Include box changes. Paths survive the tree being built again; item objects do not.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

7.2.75 `byPath(path)` EXPORTED line 417

Finds the item with a path in the tree as it is now, after it was built again.

Called by in js/tests/behaviour_test.js: `beyond()`; in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

7.2.76 `onTop(item)` INTERNAL line 419

Tells whether an item is the top of its line: a unit or technology right under a building or group. The tests work this out on their own, apart from the page's code (Req. 27).

Called by `inner()`, `fixtures()`
Calls `itemAbove()`

7.2.77 `lines()` INTERNAL line 421

Lists the items that start a line.

Called by `fixtures()`
Calls in js/model/model.js: `lineOf()`

7.2.78 `inner(item)` INTERNAL line 423

Tells whether an item is a unit or technology below the top of its line with one item under it, where a mark passes down only at the same count (Req. 29, 30).

Called by `fixtures()`
Calls `itemAbove()`, `onTop()`

7.2.79 `fixtures()` EXPORTED lines 425–459

Picks sample items from the tree: leaves some civilisations have, buildings, line tops, even and uneven counts, a line in a group of lines, a building some do without, a lone building and an item with a rival. Found from the data, to survive game updates.

Called by `tipGaps()`; in js/tests/run_test.js: `rounds()`; in js/tests/behaviour_test.js: `standIns()`, `lines()`; in js/tests/interface_test.js: `accessibilityTests()`; in js/tests/pictures_test.js: `pictureTests()`
Calls `itemAbove()`, `lines()`, `onTop()`, `inner()`; in js/model/datahandler.js: `size()`, `nameOf()`; in js/model/model.js: `whole()`, `lineOf()`, `lineEnd()`

7.3 static_test.js

Folder: js/tests/ 385 lines 22 functions, 7 exported
Imports: js/model/datahandler.js, js/model/model.js, js/tests/helpers_test.js
Imported by: js/tests/run_test.js

The maintainability and data tests: they read the page's own files as text and check the rules, then check data.json and the updater's writing.

7.3.1 `scripts()` EXPORTED lines 11–36

Reads every script reachable from `js/script.js`, following its imports, and notes which ones load only through a later import, as the tests do.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `load()`

7.3.2 `styles()` EXPORTED lines 38–39

Reads every style sheet that index.html links, with its address.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `load()`

7.3.3 `linesOfCode(code)` INTERNAL line 46

Counts the lines that are not blank, as Rules 23 and 25 count a function or a file.

Called by `longFunctions()`, `sizeProblems()`
Calls *no function of the code*

7.3.4 `regexMayStart(last, word)` INTERNAL lines 48–49

Tells whether a / after a given character or word starts a regular expression rather than a division, so `blanked()` skips the right text.

Called by `blanked()`
Calls *no function of the code*

7.3.5 `closing(code, at)` INTERNAL lines 51–62

Finds where a string or regular expression that starts at a place ends, minding escapes and, for a regular expression, character sets.

Called by `blanked()`
Calls *no function of the code*

7.3.6 `blanked(code)` INTERNAL lines 64–106

Blanks out the insides of strings, regular expressions and template texts, keeping the code inside `${}` and every line break, so that only real braces and parentheses are left to count.

Called by `functionsOf()`
Calls `regexMayStart()`, `closing()`

7.3.7 `opensBody(text, at, paren)` INTERNAL lines 108–114

Tells whether a brace opens a function's body: after =>, or after a parameter list that follows a name which is not `if`, `for`, `while`, `switch`, `catch` or `with`.

Called by `functionsOf()`
Calls *no function of the code*

7.3.8 `functionsOf(code)` INTERNAL lines 116–132

Finds where every function body in a script starts and ends, by pairing its braces in the blanked text.

Called by `longFunctions()`
Calls `blanked()`, `opensBody()`

7.3.9 `longFunctions(code, lines)` INTERNAL lines 134–144

Names every function over 60 lines of code, with its line (Rule 23).

Called by `sizeProblems()`
Calls `linesOfCode()`, `functionsOf()`

7.3.10 `sizeProblems(page, sheets, files)` INTERNAL lines 146–165

Lists every file over 400 lines of code, every function over 60 and every script line over 120 characters (Rules 23 to 25).

Called by `sourceTests()`
Calls `linesOfCode()`, `longFunctions()`

7.3.11 `unspun(sheet)` INTERNAL line 177

Takes the loading picture's spin out of the style sheets' text, the only animation allowed, before they are searched for others.

Called by `sourceTests()`
Calls *no function of the code*

7.3.12 `badImages(page, sheets)` INTERNAL lines 179–193

Tells whether any picture breaks the rules: media elements, images outside templates or with an address in them, a picture in CSS outside a `.pictures` rule or outside `assets/images/`, or a font outside `assets/fonts/` (Rules 9, 22).

Called by `sourceTests()`
Calls *no function of the code*

7.3.13 `mvcProblems(files)` INTERNAL lines 201–229

Lists what breaks model, view and controller: a model script importing a view or a controller, a view importing a controller, saving a setting or naming a mark in a call to its controller, a controller touching the page or the address, a script outside js/model/ and the tests working on the model's sets, anything but a test calling `page.update()`, a model.js that does not share `subscribe` and `unsubscribe`, shares `notify` or holds texts, another script than page.js observing the model, and, once the tests have run, an item holding a page element (Test 85).

Called by `sourceTests()`
Calls in js/tests/helpers_test.js: `need()`

7.3.14 `sourceTests(page, sheets, files)` EXPORTED lines 231–282

The rules tests: no comments; no CSS or HTML in the scripts, which hand CSS only numbers; where the files are and what they import, the tests named `_test.js` and loading only when testing, and no two of the files with one name (Rule 29); the size limits; which scripts touch the page, store data or write the address, and model, view and controller kept apart (`mvcProblems()`); images, fonts, sounds, animations and dependencies; and no game data in the scripts.

Called by in js/tests/run_test.js: `rounds()`
Calls `sizeProblems()`, `mvcProblems()`, `badImages()`, `unspun()`; in js/tests/helpers_test.js: `expect()`, `keys()`

7.3.15 `textTests()` EXPORTED lines 284–292

Checks that every language has all of the page's text, and names every item and civilisation.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `missingText()`; in js/tests/helpers_test.js: `keys()`, `expect()`, `codes()`

7.3.16 `tagTests()` EXPORTED lines 294–297

Checks that every language has a name and a valid language tag (WCAG 3.1.1).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/tests/helpers_test.js: `expect()`, `codes()`

7.3.17 `alike(a, b)` INTERNAL line 301

Tells whether two values are equal, by their JSON.

Called by `badExtras()`
Calls *no function of the code*

7.3.18 `badExtras(all)` INTERNAL lines 303–315

Lists the entries of `extras` that are wrong: not in the tree, without a name number or a list of known civilisations, or whose availability and icon in data.json do not match what `fromExtra()` makes of them.

Called by `treeTests()`
Calls `alike()`; in js/model/datahandler.js: `fromExtra()`, `availabilityOf()`, `pictureOf()`

7.3.19 `treeTests(all)` INTERNAL lines 317–340

Checks the tree in data.json: every item once, only known keys, flags only where they belong, `"excludes"` and `"replaces"` naming other known items, and every item with its civilisations and its icon.

Called by `dataTests()`
Calls `badExtras()`; in js/tests/helpers_test.js: `expect()`

7.3.20 `textDataTests(all)` INTERNAL lines 342–361

Checks the texts in data.json: counted texts with an "other" form, the same placeholders in every language, overrides of known names, and the source, tool texts and ignore list filled in.

Called by `dataTests()`
Calls in js/tests/helpers_test.js: `codes()`, `expect()`

7.3.21 `dataTests()` EXPORTED lines 363–367

Runs the checks of the tree and of the texts in data.json.

Called by in js/tests/run_test.js: `rounds()`
Calls `treeTests()`, `textDataTests()`; in js/tests/helpers_test.js: `keys()`

7.3.22 `updaterTests()` EXPORTED lines 369–385

Checks the updater's writing without going online: the files it writes read back the same, availability is stored in its shortest form, and stored names are already tidy.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `pack()`, `toJson()`, `unpack()`, `addLanguage()`, `localeFile()`, `availabilityOf()`, `civsOf()`, `setOf()`, `joinLines()`; in js/tests/helpers_test.js: `codes()`, `check()`, `same()`, `expect()`, `keys()`

7.4 behaviour_test.js

Folder: js/tests/ 430 lines 17 functions, 1 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/controller/optionshandler.js, js/tests/helpers_test.js

Imported by: js/tests/run_test.js

The tests of the tree: marking, Filters, the counts, opening and closing, and Find. Each test function names its checks after TESTS.md, which says on which lines each test is found.

7.4.1 `sameCivs(item)` INTERNAL

lines 11–15

Lists the items of a line, from an item down while each has one item under it that is not a group, that have the same civilisations as the item: those a mark on it passes down to (Req. 29, 30).

Called by `lineTops()`, `lines()`
Calls *no function of the code*

7.4.2 `chipNames()` INTERNAL

line 17

Lists the names on the Filters buttons.

Called by `branches()`, `lineTops()`, `rivals()`, `standIns()`
Calls in js/view/stringbuilder.js: `$()`

7.4.3 `fullName(item)` INTERNAL

line 19

Returns "Full" and an item's name, as Filters names a fully marked building (Req. 24).

Called by `branches()`
Calls in js/model/datahandler.js: `fill()`, `nameOf()`

7.4.4 `marking({ partial, other, third })` INTERNAL

lines 21–39

Tests marking by click, X, the menu key and Space, the civilisation list and its count, and the first Filters button (Tests 3 to 7; Req. 3, 4, 22).

Called by `behaviourTests()`
Calls in js/model/datahandler.js: `plural()`, `size()`, `fill()`; in js/view/stringbuilder.js: `$()`, `chipName()`; in js/view/treeview.js: `rowOf()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `check()`, `same()`, `listed()`, `expected()`, `keyDown()`

7.4.5 `branches({ branch, line, group, headed, lone, walled })` INTERNAL

lines 41–89

Tests buildings, groups and lines: what marking them marks and shows, their Filters buttons and names, "Full" only with something under it and not in a group of lines, and the tick, cross and dash (Tests 8 to 14; Req. 14, 23 to 26, 30).

Called by `behaviourTests()`
Calls `chipNames()`, `fullName()`; in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `state()`, `lineOf()`, `lineEnd()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `setOpen()`, `shown()`, `rowOf()`; in js/tests/helpers_test.js: `reset()`, `below()`, `tap()`, `nodeOf()`, `check()`, `same()`, `drawn()`, `keyDown()`

7.4.6 `crowding()` INTERNAL

lines 91–128

Tests that a full tree and the longest civilisation list never make the page scroll, crush a pane or overlap, in whatever window the tests run (Test 15; Rule 12).

Called by `behaviourTests()`
Calls in js/model/datahandler.js: `size()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `listed()`, `expect()`

7.4.7 `gestures({ third })` INTERNAL

lines 130–147

Tests that a long press with a finger marks must lack, and that a right-click clears it (Test 16; Req. 3).

Called by `behaviourTests()`
Calls in js/tests/helpers_test.js: `reset()`, `nodeOf()`, `wait()`, `check()`

7.4.8 `lineTops({ line })` INTERNAL

lines 149–171

Tests that marking the top of a line passes the mark down only at the same count, must have or must lack, that marking it again clears it, and that marking a line's last item marks the line before it with one filter (Tests 17, 18; Req. 29, 30).

Called by `behaviourTests()`

Calls `sameCivs()`, `chipNames()`; in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `lineOf()`, `lineEnd()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `same()`, `listed()`, `expected()`, `keyDown()`, `check()`

7.4.9 `parents({ branch, loose, without })` INTERNAL

lines 173–187

Tests that must have passes up but never to a building, which then shows a dash, and that an item can be must lack while everything under it is must have (Tests 19, 20; Req. 30).

Called by `behaviourTests()`

Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `below()`, `tap()`, `nodeOf()`, `check()`, `keyDown()`, `listed()`, `same()`, `expected()`

7.4.10 `rivals({ rival })` INTERNAL

lines 189–230

Tests an item that excludes another, such as the Xolotl Warrior and the Stable: must have on the one greys out the other and everything in it, the one staying in sight and the other's arrow working; must have on any of those greys out the one; what is greyed out cannot be marked; and must lack greys out nothing (Test 21).

Called by `behaviourTests()`

Calls `chipNames()`; in js/model/model.js: `state()`; in js/view/stringbuilder.js: `$()`, `chipName()`; in js/view/treeview.js: `setOpen()`, `shown()`, `isOpen()`; in js/tests/helpers_test.js: `reset()`, `check()`, `nodeOf()`, `tap()`, `same()`, `listed()`, `expected()`, `twistyOf()`, `keyDown()`

7.4.11 `standIns(sample)` INTERNAL

lines 232–293

Tests an item that replaces others, such as the Pasture and the Farm: it holds everything in their buildings but their unique items and groups, as the same items, so marking one marks both with one filter; must have on one greys out and closes every alternative until it is cleared (Test 22; Req. 37).

Called by `behaviourTests()`

Calls `chipNames()`; in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `isUnitOrTech()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `setOpen()`, `isOpen()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `check()`, `nodeOf()`, `below()`, `tap()`, `keyDown()`, `twistyOf()`, `same()`, `ownKeys()`, `alternativesOf()`, `fixtures()`

7.4.12 `beyond()` INTERNAL

lines 295–311

With Include ticked, finds an item that some civilisations have without the unit or technology above it, checks that must have on it and must lack on the one above list just those civilisations, and that unticking Include lifts it up with its mark kept. Part of Test 24.

Called by `lines()`

Calls in js/model/datahandler.js: `size()`; in js/model/model.js: `isUnitOrTech()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `setOpen()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `keyDown()`, `same()`, `listed()`, `expected()`, `byPath()`

7.4.13 `lines(sample)` INTERNAL

lines 313–340

Tests that marking a unit or technology opens it to its first fork or its end, and that below the top of a line a mark passes down only while the civilisations stay the same (Tests 23, 24; Req. 28 to 31).

Called by `behaviourTests()`

Calls `sameCivs()`, `beyond()`; in js/model/model.js: `lineOf()`, `isUnitOrTech()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `setOpen()`, `isOpen()`, `shown()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `check()`, `twistyOf()`, `keyDown()`, `same()`, `listed()`, `expected()`, `fixtures()`

7.4.14 `counts({ partial, other, upgraded })` INTERNAL

lines 342–376

Tests the numbers on the items: closed and open, on lines, and with filters (Tests 25 to 28; Req. 3).

Called by `behaviourTests()`

Calls in js/model/datahandler.js: `fill()`, `size()`, `plural()`, `nameOf()`; in js/model/model.js: `lineEnd()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `isOpen()`, `rowOf()`; in js/tests/helpers_test.js: `reset()`, `expect()`, `check()`, `tap()`, `nodeOf()`, `keyDown()`

7.4.15 `tree()` INTERNAL

lines 378–396

Tests Expand all, Collapse all and the arrows, and that opening a branch also opens its only children (Tests 29, 30; Req. 18, 19).

Called by `behaviourTests()`

Calls in js/model/datahandler.js: `nameOf()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `isOpen()`, `rowOf()`; in js/tests/helpers_test.js: `reset()`, `twistyOf()`, `check()`, `expect()`, `chainOf()`

7.4.16 `finding({ partial, rival })` INTERNAL

lines 398–415

Tests that Find highlights its matches in yellow, opens the branches down to them and counts them beside the box, drops a match once a mark greys out its branch (`dropsGreyed()`), and says when nothing matches (Tests 31, 32; Req. 20, 21).

Called by `behaviourTests()`

Calls in js/model/datahandler.js: `nameOf()`, `plain()`, `plural()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `shown()`; in js/tests/helpers_test.js: `reset()`, `type()`, `nodeOf()`, `yellow()`, `beside()`, `dropsGreyed()`, `check()`

7.4.17 `behaviourTests(sample)` EXPORTED

lines 417–430

Runs the tests of the tree, in the order TESTS.md lists them.

Called by in js/tests/run_test.js: `rounds()`

Calls `marking()`, `branches()`, `crowding()`, `gestures()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`, `finding()`

7.5 interface_test.js

Folder: js/tests/ 417 lines 17 functions, 8 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/optionsview.js, js/view/report.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/audit_test.js
Imported by: js/tests/run_test.js

The tests of the language menu, the languages, the Options window, the keyboard and the report window, and the runs of the accessibility audit.

7.5.1 `menuTests({ partial })` EXPORTED lines 17–63

Tests the language menu: in Options, every language in its own language, closing on Escape, a click or leaving it, a window that never resizes, the language changed on OK and kept in the address, English first, and marks kept (Tests 33 to 37; Req. 8 to 12).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `options()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `box()`, `check()`, `codes()`, `keyDown()`, `nameFor()`, `ui()`, `englishFirst()`, `same()`, `listed()`, `expected()`, `pick()`

7.5.2 `languageTests(code, { partial, other })` EXPORTED lines 65–92

Tests one language: the page's text, the tree's names, no placeholders, the footer's links and Microsoft's notice, and a translated, sorted civilisation list (Tests 38 to 42; Req. 4, 5, 11).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `nameOf()`, `fill()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `pick()`, `check()`, `labelOf()`, `visibleTexts()`, `tidy()`, `spans()`, `tap()`, `nodeOf()`, `keyDown()`, `same()`, `listed()`, `expected()`

7.5.3 `optionTests()` EXPORTED lines 94–151

Tests the two Include boxes: what the tree leaves out by default, ticking and unticking with marks kept, the choices remembered, and Options opening and closing properly (Tests 43 to 46; Req. 7, 13 to 16).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `size()`, `recall()`, `attempt()`; in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `options()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `keys()`, `uniqueKeys()`, `regionalKeys()`, `check()`, `shownKeys()`, `keptPath()`, `tap()`, `nodeOf()`, `byPath()`, `listed()`, `same()`, `rendered()`, `outside()`

7.5.4 `menuAudit({ partial })` INTERNAL lines 153–160

Runs the accessibility audit with a mark and the language menu open. This is the audit's first run, so its checks are listed from here.

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`; in js/tests/audit_test.js: `audit()`

7.5.5 `menuKeys()` INTERNAL lines 162–177

Tests the language menu's keys: Down opens it at the current language, and Down and End move through it (Test 78; WCAG 2.1.1).

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `options()`; in js/tests/helpers_test.js: `reset()`, `keyDown()`, `codes()`, `check()`

7.5.6 `languageAudits(code, { partial, other })` INTERNAL lines 179–191

Runs the accessibility audit in one language: as loaded, with filters, and with everything open. The focus checks run only in the first language.

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `pick()`, `codes()`, `tap()`, `nodeOf()`, `keyDown()`; in js/tests/audit_test.js: `audit()`

7.5.7 `spacingTests({ partial })` INTERNAL lines 193–203

Tests that wider text spacing, as WCAG 1.4.12 describes it, makes nothing stick out sideways (Test 79).

Called by `accessibilityTests()`
Calls in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `expect()`, `everything()`, `box()`, `describe()`

7.5.8 `optionKeys()` INTERNAL lines 205–229

Tests that Space ticks and unticks Include and Include Regionals, and that OK then shows or leaves out those items, keeping marks (Test 80; WCAG 2.1.1).

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `uniqueKeys()`, `regionalKeys()`, `keptPath()`, `tap()`, `nodeOf()`, `byPath()`, `listed()`, `keyDown()`, `shownKeys()`, `check()`, `same()`

7.5.9 `optionAudits()` INTERNAL lines 231–241

Runs the audit with the Options window open, and with everything included and open.

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`; in js/tests/audit_test.js: `audit()`

7.5.10 `filterFocus({ other, third })` INTERNAL lines 243–259

Tests that a Filters button clears its item and hands focus to the next button, and that clearing the last one hands it to the civilisations' heading, with every civilisation listed (Tests 56, 57).

Called by `keyboardTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `keyDown()`, `nodeOf()`, `check()`, `same()`, `listed()`, `expected()`

7.5.11 `keyboard({ line })` INTERNAL lines 261–290

Tests the arrow keys in the tree: Right opens a line all the way down, or opens a branch and goes into it, Left goes back out and closes it, Down and Up step between items, End and Home go to the last and the first, and Down on an arrow goes to the next arrow (Tests 58 to 61; WCAG 2.1.1).

Called by `keyboardTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `isOpen()`; in js/tests/helpers_test.js: `reset()`, `keyDown()`, `nodeOf()`, `check()`, `chainOf()`, `twistyOf()`

7.5.12 `findKey({ partial })` INTERNAL lines 292–299

Tests that Enter in Find goes to the first match (Test 62).

Called by `keyboardTests()`
Calls in js/model/datahandler.js: `nameOf()`, `plain()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `type()`, `keyDown()`, `check()`, `nodeOf()`

7.5.13 `keyboardTests(sample)` EXPORTED lines 301–305

Runs the keyboard tests: `filterFocus()`, `keyboard()` and `findKey()`.

Called by in js/tests/run_test.js: `rounds()`
Calls `filterFocus()`, `keyboard()`, `findKey()`

7.5.14 `accessibilityTests(sample)` EXPORTED lines 307–317

Runs this file's audits and accessibility tests in the order TESTS.md lists them: the menu, the first language, spacing and keys, then every other language and the Options window.

Called by in js/tests/run_test.js: `rounds()`
Calls `menuAudit()`, `menuKeys()`, `languageAudits()`, `spacingTests()`, `optionKeys()`, `optionAudits()`; in js/tests/helpers_test.js: `codes()`, `fixtures()`

7.5.15 `reportTests()` EXPORTED lines 319–346

Tests the report window with 200 lines in it: its title, its count and Close stay in view and only its list scrolls, and Close has focus, sits on the right and is the plain button, with or without Simple UI (Test 98).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/report.js: `openReport()`, `report()`; in js/controller/optionshandler.js: `setSimple()`; in js/tests/helpers_test.js: `box()`, `check()`

7.5.16 `uniqueFilterTests()` EXPORTED

lines 348–386

Tests that a unique item marked must have gets its own Filters button, even in a full group, which clears only it, while its line's or group's button leaves it marked, and that a wholly unique line has one button (Test 53; Req. 36).

Called by in js/tests/run_test.js: `rounds()`

Calls in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `lineOf()`, `lineEnd()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `setOpen()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `check()`, `tap()`, `nodeOf()`, `same()`

7.5.17 `headingTests()` EXPORTED

lines 388–417

Tests a group inside a building, such as the Castle's Unique Units: it has no checkbox, a click on its name or arrow opens and closes it, a right-click marks nothing, and Down from the row above reaches its arrow (Test 54).

Called by in js/tests/run_test.js: `rounds()`

Calls in js/view/treeview.js: `setOpen()`, `isOpen()`, `shown()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `check()`, `nodeOf()`, `labelOf()`, `twistyOf()`, `keyDown()`

7.6 pictures_test.js

Folder: js/tests/ 345 lines 20 functions, 1 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/optionsview.js, js/view/scrollbars.js, js/controller/optionshandler.js, js/tests/helpers_test.js
Imported by: js/tests/run_test.js

The tests of the pictures, run with Simple UI off. There the page need not meet WCAG (Rule 22), nor the UI particulars of REQUIREMENTS.md (Req. 0).

7.6.1 `at(url, path)` INTERNAL — line 16

Tells whether a CSS `url()` points at a given picture in `assets/images/`.

Called by `picture()`, `findLooks()`
Calls *no function of the code*

7.6.2 `picture(style, path)` INTERNAL — line 18

Tells whether an element's background is a given picture.

Called by `cut()`, `wrongSizes()`, `pictureLooks()`, `wrongBoxes()`, `offCell()`, `findLooks()`
Calls `at()`

7.6.3 `cut(el)` INTERNAL — lines 20–23

Tells whether a button is not drawn as the whole button picture: the wrong file or sheet size, drawn parts left on, or not 290 by 50 in proportion.

Called by `toolLooks()`
Calls `picture()`; in js/tests/helpers_test.js: `grade()`, `styles()`, `box()`

7.6.4 `offParchment(style)` INTERNAL — lines 25–43

Lists the header, panes and footer that stick out of the parchment, worked out from the five numbers `--ratio`, `--left`, `--top`, `--right` and `--bottom` in pictures.css and the window's size.

Called by `pictureLooks()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `box()`, `rendered()`, `describe()`

7.6.5 `strayCursors()` INTERNAL — lines 45–52

Lists the elements without the right cursor: the pointing hand over links and the sword elsewhere, `_x1` at their own size or `_x2` at two thirds, at 1.5x (Test 47).

Called by `pictureLooks()`
Calls in js/tests/helpers_test.js: `grade()`, `everything()`, `describe()`

7.6.6 `wrongSizes()` INTERNAL — lines 54–67

Lists the pictures that do not suit the parchment chosen, trying every parchment in turn: `_sd` and `_x1` pictures with fhd and qhd, `_4k` and `_x2` ones with uhd and ultrawide.

Called by `pictureLooks()`
Calls `picture()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `everything()`, `styles()`, `urlsOf()`, `fitting()`, `need()`

7.6.7 `pictureLooks()` INTERNAL — lines 69–87

Tests the pictures as a whole: the choice for each screen, background, cursors, the Options button and its ring, the parchment and the footer (Test 47).

Called by `pictureTests()`
Calls `picture()`, `wrongSizes()`, `offParchment()`, `strayCursors()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `fitted()`, `expect()`, `need()`, `misjudged()`, `grade()`, `spans()`

7.6.8 `wrongBoxes()` INTERNAL — lines 89–105

Lists the checkboxes without the sheet's picture for their state, empty, ticked or crossed, or with a drawn mark other than the dash, the radio buttons without theirs, and any state with no box (Test 48).

Called by `pictureTests()`
Calls `picture()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `grade()`, `drawn()`, `describe()`

7.6.9 `cellIn(names, name, side)` INTERNAL lines 107–112

Works out where a picture's cell is in its atlas: the files sorted by name, row by row, in a square grid.

Called by `unpictured()`
Calls *no function of the code*

7.6.10 `share(part, of)` INTERNAL line 114

Turns a column or row into the percentage CSS needs to show that cell of an atlas.

Called by `offCell()`
Calls *no function of the code*

7.6.11 `offCell(icon, cell, path, grids)` INTERNAL lines 116–124

Tells whether an icon does not show its cell of its atlas: the wrong file, repeat, box, size or position. Notes the atlas for `misfit()`.

Called by `unpictured()`
Calls `picture()`, `share()`

7.6.12 `misfit(grids)` INTERNAL lines 126–131

Lists the atlases whose size does not fit their grid of 48 px cells, which would shift every icon.

Called by `unpictured()`
Calls *no function of the code*

7.6.13 `unpictured()` INTERNAL lines 133–158

With every item included, lists the tree rows without the right icon: its cell of its atlas, 44 px, between box and name, and outlined as `colour_key.csv` says (Test 48).

Called by `pictureTests()`
Calls `offCell()`, `cellIn()`, `misfit()`; in js/model/datahandler.js: `civFile()`; in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `edges()`, `nodeOf()`, `labelOf()`, `grade()`, `need()`

7.6.14 `findLooks()` INTERNAL lines 160–179

Tests the Find box with the pictures: its frame, search icon and clearing cross, its width of up to 325 px, the message beside it, and that matches are marked (Test 49; Req. 21).

Called by `pictureTests()`
Calls `at()`, `picture()`; in js/model/datahandler.js: `nameOf()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `box()`, `rendered()`, `type()`, `beside()`

7.6.15 `toolLooks()` INTERNAL lines 181–226

Lists what is wrong with the tree's buttons: not Clear and Expand all, or Collapse all while a branch is open, not side by side as wide as the pane, not the whole button picture, text cut, moved, focus not handed over, or different in another language (Test 50).

Called by `pictureTests()`
Calls `cut()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `setOpen()`; in js/tests/helpers_test.js: `box()`, `rendered()`, `need()`, `describe()`, `same()`, `tap()`, `nodeOf()`, `twistyOf()`, `codes()`, `pick()`

7.6.16 `simpleSwitch()` INTERNAL lines 228–261

Ticks and unticks Simple UI, and lists what went wrong: a picture or link left, the box moved, the choice not remembered, or something not back (Test 51).

Called by `pictureTests()`
Calls in js/model/datahandler.js: `size()`, `recall()`, `attempt()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `pictures()`, `simpleSpot()`, `picturesOn()`, `civLinks()`, `simpleTicked()`, `same()`, `rendered()`, `need()`, `ui()`

7.6.17 `strayBars(when)` INTERNAL lines 263–272

Lists the lists whose scroll bar is wrong: one shown where nothing scrolls or with Simple UI, none where the list scrolls, or one that `barProblems()` finds fault with.

Called by `barLooks()`
Calls in js/view/stringbuilder.js: `simple()`; in js/tests/helpers_test.js: `painted()`, `rendered()`, `need()`, `describe()`, `barProblems()`

7.6.18 `ownBars(width)` INTERNAL

lines 274–275

Lists the lists whose own browser scroll bar is not as wanted: hidden with the pictures and back with Simple UI.

Called by `barLooks()`
Calls in js/tests/helpers_test.js: `describe()`

7.6.19 `barLooks()` INTERNAL

lines 277–307

Tests the pictures' scroll bars with every branch closed and open, scrolled to the end, in Options, with the language list open, whose wheel scrolls a language at a time, and with Simple UI, and that the down arrow and the wheel scroll (Test 52).

Called by `pictureTests()`
Calls `ownBars()`, `strayBars()`; in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `setMenu()`, `options()`; in js/controller/optionshandler.js: `setSimple()`; in js/tests/helpers_test.js: `box()`, `tap()`, `nodeOf()`, `need()`, `reset()`

7.6.20 `pictureTests()` EXPORTED

lines 309–344

Runs the tests of the pictures with Simple UI off, then turns it on again (Tests 47 to 52; Rules 9, 18, 22; Req. 0, 32, 33).

Called by in js/tests/run_test.js: `rounds()`
Calls `pictureLooks()`, `wrongBoxes()`, `unpictured()`, `findLooks()`, `toolLooks()`, `simpleSwitch()`, `barLooks()`; in js/controller/optionshandler.js: `setSimple()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `keyDown()`, `check()`, `tipLooks()`, `tipGaps()`, `expect()`

7.7 audit_test.js

Folder: js/tests/ 181 lines 14 functions, 1 exported

Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/treeview.js, js/view/optionsview.js, js/tests/helpers_test.js

Imported by: js/tests/interface_test.js

The accessibility audit: checks the page as it stands against WCAG 2.2 AAA, wherever a test calls it. Its checks are Tests 63 to 77.

7.7.1 `rgba(value)` INTERNAL

lines 13–16

Reads a CSS colour into its red, green, blue and alpha numbers.

Called by `backgroundOf()`, `contrastOn()`
Calls *no function of the code*

7.7.2 `blend([r, g, b, a], [R, G, B])` INTERNAL

line 18

Blends a see-through colour over the colour below it, to get the colour actually seen.

Called by `backgroundOf()`, `contrastOn()`
Calls *no function of the code*

7.7.3 `backgroundOf(el)` INTERNAL

lines 20–28

Works out the colour behind an element by blending its own and its ancestors' backgrounds. Contrast is measured against this.

Called by `colourChecks()`
Calls `rgba()`, `blend()`

7.7.4 `luminance(rgb)` INTERNAL

lines 30–36

Computes a colour's relative luminance, as WCAG defines it.

Called by `contrast()`
Calls *no function of the code*

7.7.5 `contrast(a, b)` INTERNAL

lines 38–41

Computes the contrast ratio of two colours, as WCAG defines it, from 1:1 to 21:1.

Called by `contrastOn()`
Calls `luminance()`

7.7.6 `contrastOn(colour, behind)` INTERNAL

line 43

Computes the contrast of a colour, which may be see-through, over the colour behind it.

Called by `colourChecks()`
Calls `contrast()`, `blend()`, `rgba()`

7.7.7 `visibleCentre(el)` INTERNAL

lines 45–58

Finds the middle of the visible part of an element's first line, clipped by every scrolling area around it, to see what is drawn there.

Called by `focusChecks()`
Calls *no function of the code*

7.7.8 `focusChecks(list)` INTERNAL

lines 60–75

Focuses every control in turn, and checks that it shows a thick outline with 3:1 contrast and that nothing covers it (Test 77; WCAG 2.4.7, 2.4.12, 2.4.13).

Called by `audit()`
Calls `visibleCentre()`; in js/tests/helpers_test.js: `focusables()`, `describe()`

7.7.9 `structureChecks(note, list)` INTERNAL

lines 77–100

Checks the page's title and languages, its landmarks and headings, and that ids are unique and ARIA references resolve (Tests 63 to 67; WCAG 1.3.1, 2.4.2, 3.1.1, 3.1.2, 4.1.2).

Called by `audit()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/optionsview.js: `options()`; in js/tests/helpers_test.js: `describe()`, `rendered()`

7.7.10 `controlChecks(list)` INTERNAL lines 102–117

Checks that boxes, branches and the language menu say their state, and that every control has a name containing its visible text (Tests 68, 69; WCAG 2.5.3, 4.1.2).

Called by `audit()`
Calls in js/model/datahandler.js: `nameOf()`; in js/view/stringbuilder.js: `$()`; in js/view/treeview.js: `rowOf()`; in js/tests/helpers_test.js: `describe()`, `twistyOf()`, `focusables()`, `rendered()`, `nameFor()`, `visibleTexts()`, `squash()`

7.7.11 `colourChecks(all, list)` INTERNAL lines 119–135

Checks the contrast of all visible text, 7:1, or 4.5:1 when large, and of control borders, 3:1 (Tests 70, 71; WCAG 1.4.6, 1.4.11).

Called by `audit()`
Calls `contrastOn()`, `backgroundOf()`; in js/tests/helpers_test.js: `describe()`

7.7.12 `layoutChecks(all, note, list)` INTERNAL lines 137–156

Checks 44 by 44 targets, no page scrolling, nothing sticking out sideways, and unjustified text with 1.5 line spacing (Tests 72 to 75; WCAG 2.5.5, 1.4.10, 1.4.8; Rule 12).

Called by `audit()`
Calls in js/tests/helpers_test.js: `tidy()`, `focusables()`, `describe()`

7.7.13 `mediaChecks(all, list)` INTERNAL lines 158–167

Checks that the page shows no media or CSS picture, and runs no animations or transitions (Test 76; Rules 9, 10, 11, 22).

Called by `audit()`
Calls in js/tests/helpers_test.js: `pictured()`, `describe()`, `styles()`

7.7.14 `audit(label, { focus = true } = {})` EXPORTED lines 169–181

Checks the page as it stands through the checks above, and the focus checks if asked, under a name that says the state. Called in many states and languages.

Called by in js/tests/interface_test.js: `menuAudit()`, `languageAudits()`, `optionAudits()`
Calls `structureChecks()`, `controlChecks()`, `colourChecks()`, `layoutChecks()`, `mediaChecks()`, `focusChecks()`; in js/tests/helpers_test.js: `check()`, `everything()`

8 locales/

1 file 609 lines

One file for each language but English, all laid out alike, so only de.json is described here.

Subchapters: 8.1 de.json

8.1 de.json

Folder: locales/ 609 lines 2 keys

The German texts. Every other language has a file like it in `locales/`, which the page loads when that language is chosen (`localeFile()`) and the updater writes. It holds the two keys of data.json that differ by language, with the same keys inside.

Lines	Key	What is there
2–6	`overrides`	German names to use instead of the game's own.
7–608	`text`	The page's texts in German, in the same four parts as data.json's `text`.
8–70	`ui`	The page's own texts.
71–77	`groups`	The name of each group.
78–135	`civs`	The name of each civilisation.
136–607	`items`	The name of each item.

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